SUPER APP

Entity Relationship Design (ERD)

Every table, column, relationship, constraint and index of the Super App database — SQL Server, EF Core Code-First.

Document	ERD — backend phase
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Audience	Backend engineers, database reviewers

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1. How to read this document

1.1 Approach

- **Code-First:** each table below becomes an entity class (`Domain/Entities`) and a Fluent API configuration class (`Infrastructure/Persistence/Configurations`). The owner runs `Add-Migration InitialCreate` and `Update-Database` .
- **Creation order** follows the rule *parent before child*: **Part A** lookup and independent tables (no foreign keys), **Part B** child tables (each one after every table it references), **Part C** junction tables (many-to-many).
- Every table states its **five constraints** where they apply: `NOT NULL` , `UNIQUE` , `DEFAULT` , `CHECK` , `FOREIGN KEY` with its `ON DELETE` behaviour.
- After the migration, open **SSMS** → **Database Diagrams** → **New Database Diagram**, add the tables of one schema at a time, and compare with the Mermaid diagrams in section 9.

1.2 Schemas

Tables are grouped in SQL Server schemas so the database is easy to browse:

Schema	Holds
`auth`	Users, roles, permissions, Identity tables, applications, ID documents
`geo`	Governorates and areas
`ops`	Settings, feature flags, integrations, webhooks, outbox, audit log, files, org chart, goals, budgets
`ai`	AI agents, their keys, tasks and runs
`pitch`	Partners, venues, pitches, hours, closures, packages options, offers, reviews
`booking`	Bookings, slots, packages, QR scans, waitlist, games, free slots, alerts, tournaments
`shop`	Stores, products, variants, carts, orders, fulfilments
`pod`	Seller levels, sellers, catalogue, designs, reports, strikes, samples
`factory`	Factories, staff, devices, prices, jobs
`ship`	Shipments, courier events, returns
`money`	Wallets, ledger, payments, refunds, payouts, approvals, receipts, promo codes, points, vouchers
`fantasy`	Competitions, seasons, clubs, players, gameweeks, fixtures, stats, teams, picks, groups, passes
`community`	Communities, posts, polls, comments, reactions, reports, banned words
`support`	Tickets, chat messages, attachments
`content`	Banners, pop-ups, news, legal documents and acceptances, help center, cookie consents
`notify`	Notifications, preferences, email outbox, broadcasts

1.3 Conventions

Topic	Rule
Names	Tables plural PascalCase (`Bookings`), columns PascalCase, foreign keys `<Parent>Id`
Primary keys	`Id bigint IDENTITY(1,1)` for business tables; `Id int` for lookups; `Id uniqueidentifier` for `auth.Users` and `auth.Roles` (ASP.NET Identity)
Public codes	Things customers see (booking, order, ticket, return, receipt) also have a short unique `Code` (e.g. `SA-7K2M9Q`); internal ids are never shown
Money	`decimal(12,2)` , EGP. Percentages `decimal(5,2)` (e.g. 12.50 = 12.5 %)
Dates	`datetime2(0)` in **UTC**; calendar dates `date` ; hours `tinyint` 0–23 (Cairo local hour of the slot)
Status / type values	`tinyint` with a `CHECK` on the allowed values; the meaning of each value is listed in the table (and as a C# enum)
JSON	`nvarchar(max)` with `CHECK (ISJSON(col) = 1)` , only for flexible snapshots (addresses at order time, application answers, event payloads)
Text in two languages	`...Ar` and `...En` columns where Super App writes the text (legal, help, catalogue details, pop-ups)
Constraint names	`PK_Table` , `FK_Child_Parent_Column` , `UQ_Table_Columns` , `CK_Table_Rule` , `DF_Table_Column` , `IX_Table_Columns`

Common column sets (written once here, referenced as tags in each table):

Tag	Columns
[B] base	`Id bigint IDENTITY PK` · `CreatedAt datetime2(0) NOT NULL DEFAULT SYSUTCDATETIME()` · `UpdatedAt datetime2(0) NULL`
[L] lookup	`Id int IDENTITY PK · IsActive bit NOT NULL DEFAULT 1 · SortOrder int NOT NULL DEFAULT 0`
[S] soft delete	`IsDeleted bit NOT NULL DEFAULT 0` · `DeletedAt datetime2(0) NULL` — EF global query filter; unique indexes are filtered `WHERE IsDeleted = 0`
[V] concurrency	`RowVersion rowversion NOT NULL` — two people cannot overwrite each other (stock, slots, wallets, payouts)
[A] author	`CreatedById uniqueidentifier NULL` FK → `auth.Users` **NO ACTION**

1.4 Delete behaviour (ON DELETE)

Rule	Used for
NO ACTION / RESTRICT (default)	Anything with history or money: users, pitches, products, designs, orders, bookings, payments, wallets, ledger. Business records are **soft-deleted** or switched off, never hard-deleted
CASCADE	Children that cannot live without their parent: photos, opening hours, variants, specs, order items, slots, messages, attachments, poll options, picks

Rule	Used for
SET NULL	Optional links where the child must survive (e.g. a ticket's linked order)

SQL Server forbids several cascade paths into one table, so where two parents could cascade, one of them is **NO ACTION** (noted in the table).

2. Creation order

Part A — Lookup and independent tables (no foreign keys)

#	Table	Purpose
A1	`geo.Governorates`	27 governorates
A2	`auth.Permissions`	Fine-grained permission codes
A3	`auth.Roles`	Identity roles (Customer, PitchOwner, Seller, FactoryOwner, Admin, CEO, Agent, custom admin roles)
A4	`auth.Users`	Identity users — every person and service account
A5	`ops.Departments`	Company departments for the org chart
A6	`ops.Settings`	Platform settings (key/value)
A7	`ops.FeatureFlags`	Kill switches and API group switches
A8	`ops.Integrations`	External services registry
A9	`ops.WebhookSubscriptions`	Outgoing webhooks (n8n)
A10	`ops.OutboxEvents`	Events waiting to be delivered
A11	`pitch.Facilities`	Parking, showers, lights…
A12	`pitch.CancellationPolicies`	Flexible, Standard, Moderate, Strict
A13	`shop.ProductCategories`	Shop categories
A14	`pod.DesignCategories`	Design sections (football, anime…)
A15	`pod.SellerLevels`	Starter, Rising, Pro, Elite and their perks
A16	`pod.CatalogProducts`	Printable products (tee, oversized tee, hoodie)
A17	`fantasy.Competitions`	Egyptian Premier League, English Premier League
A18	`fantasy.Clubs`	Football clubs
A19	`fantasy.Sponsors`	Fantasy sponsors
A20	`content.LegalDocuments`	Terms, privacy, refunds, waiver, agreements, cookies
A21	`content.HelpTopics`	Help center topics
A22	`content.Banners`	Home banners
A23	`content.PopupAds`	Pop-up ads
A24	`community.BannedWords`	Blocked words
A25	`ops.PublicHolidays`	Egyptian public holidays (synced; Islamic dates estimated, confirmed by an admin)

Part B — Child tables (in dependency order)

#	Table	Depends on
B1	`geo.Areas`	Governorates
B2	`auth.RoleClaims`	Roles
B3	`auth.UserClaims`	Users
B4	`auth.UserLogins`	Users
B5	`auth.UserTokens`	Users
B6	`auth.StaffLimits`	Users
B7	`auth.UserAddresses`	Users, Areas
B8	`ops.Files`	Users
B9	`auth.KycDocuments`	Users, Files
B10	`auth.Applications`	Users
B11	`auth.ApplicationFiles`	Applications, Files
B12	`ops.OrgPositions`	Departments, Users, OrgPositions (self)
B13	`ai.Agents`	Users, OrgPositions
B14	`ai.ApiKeys`	Agents
B15	`ai.AgentTasks`	Agents
B16	`ai.AgentRuns`	Agents, AgentTasks
B17	`ops.AuditLogs`	Users, Agents
B18	`ops.IntegrationUsage`	Integrations
B19	`ops.WebhookDeliveries`	WebhookSubscriptions, OutboxEvents
B20	`ops.Goals`	Departments, Users
B21	`ops.KeyResults`	Goals
B22	`ops.Budgets`	Departments
B23	`ops.ComplianceItems`	Departments, Users
B24	`pitch.Partners`	Users
B25	`pitch.PartnerStaff`	Partners, Users
B26	`pitch.Venues`	Partners, Areas
B27	`pitch.Pitches`	Venues, CancellationPolicies, Users
B28	`pitch.PitchPhotos`	Pitches, Files
B29	`pitch.OpeningHours`	Pitches

#	Table	Depends on
B30	`pitch.Closures`	Pitches, Users
B31	`pitch.PackageOptions`	Pitches, Users
B32	`pitch.Offers`	Pitches
B33	`shop.Stores`	Users, Files
B34	`pod.Sellers`	Users, SellerLevels, Stores
B35	`pod.SellerStrikes`	Sellers, Users
B36	`pod.LevelReviews`	Sellers, SellerLevels
B37	`pod.CatalogColors`	CatalogProducts
B38	`pod.CatalogSizes`	CatalogProducts
B39	`pod.CatalogDetails`	CatalogProducts
B40	`pod.CatalogPrintAreas`	CatalogProducts
B41	`factory.Factories`	Users
B42	`factory.FactoryStaff`	Factories
B43	`factory.FactoryDevices`	Factories
B44	`factory.FactoryPrices`	Factories, CatalogProducts
B45	`pod.CatalogChangeRequests`	CatalogProducts, Factories, Users
B46	`pod.Designs`	Sellers, CatalogProducts, DesignCategories, Files, Users
B47	`pod.DesignTags`	Designs
B48	`pod.DesignReports`	Designs, Users
B49	`shop.Products`	Stores, ProductCategories, Users
B50	`shop.ProductImages`	Products, Files
B51	`shop.ProductVariants`	Products
B52	`shop.ProductSpecs`	Products
B53	`money.Wallets`	Users
B54	`money.PromoCodes`	Users, Stores
B55	`money.Payments`	Users
B56	`booking.Packages`	Pitches, Users, PackageOptions, Payments
B57	`booking.Series`	Pitches, Users
B58	`booking.Bookings`	Pitches, Users, Packages, Series, Payments, PromoCodes
B59	`booking.BookingSlots`	Bookings, Pitches
B60	`booking.SlotHolds`	Pitches, Users

#	Table	Depends on
B61	`booking.QrScans`	Bookings, Packages, Users
B62	`booking.WaitlistEntries`	Pitches, Users
B63	`booking.FreeSlots`	Pitches, Users, Bookings
B64	`booking.SlotAlerts`	Users, Pitches, Areas
B65	`booking.Games`	Users, Bookings
B66	`booking.GameRequests`	Games, Users, Payments
B67	`booking.PlayerRatings`	Games, Users
B68	`booking.Tournaments`	Partners, Pitches
B69	`booking.TournamentTeams`	Tournaments, Users, Payments
B70	`booking.TournamentMatches`	Tournaments, TournamentTeams
B71	`pitch.Reviews`	Pitches, Bookings, Users
B72	`pitch.ReviewReports`	Reviews, Users
B73	`shop.Orders`	Users, PromoCodes, Payments
B74	`shop.OrderItems`	Orders, ProductVariants, Designs, Stores
B75	`shop.Fulfilments`	Orders, Stores
B76	`shop.CartItems`	Users, ProductVariants, Designs
B77	`factory.Jobs`	OrderItems, Factories, FactoryStaff
B78	`factory.JobEvents`	Jobs, FactoryStaff
B79	`factory.JobProblems`	Jobs, FactoryStaff, Users
B80	`pod.Samples`	Sellers, Designs, Orders
B81	`ship.Shipments`	Orders, Fulfilments, Files
B82	`ship.ShipmentEvents`	Shipments
B83	`ship.Returns`	Orders, Users
B84	`ship.ReturnItems`	Returns, OrderItems
B85	`money.LedgerEntries`	Wallets
B86	`money.Refunds`	Payments, Users
B87	`money.PayoutRequests`	Wallets, Users
B88	`money.Approvals`	Users
B89	`money.ApprovalDecisions`	Approvals, Users
B90	`money.Adjustments`	Wallets, Users
B91	`money.Receipts`	Users, Files

#	Table	Depends on
B92	money.PromoCodeUses	PromoCodes, Users, Orders, Bookings
B93	money.PointEntries	Users
B94	money.Vouchers	Users
B95	fantasy.Seasons	Competitions
B96	fantasy.Players	Clubs
B97	fantasy.Gameweeks	Seasons
B98	fantasy.Fixtures	Gameweeks, Clubs
B99	fantasy.PlayerFixtureStats	Players, Fixtures
B100	fantasy.FantasyTeams	Users, Seasons
B101	fantasy.Picks	FantasyTeams, Gameweeks, Players
B102	fantasy.Transfers	FantasyTeams, Gameweeks, Players
B103	fantasy.TeamGameweekPoints	FantasyTeams, Gameweeks
B104	fantasy.FantasyGroups	Seasons, Users, Sponsors
B105	fantasy.FantasyPasses	Users, Competitions, Payments
B106	community.Communities	Users
B107	community.Posts	Communities, Users, Files
B108	community.PollOptions	Posts
B109	community.Comments	Posts, Comments (self), Users
B110	community.Reactions	Users, Posts, Comments
B111	community.ContentReports	Users, Posts, Comments
B112	support.Tickets	Users, Agents, Bookings, Orders
B113	support.TicketMessages	Tickets, Users
B114	support.TicketAttachments	TicketMessages, Files
B115	content.HelpArticles	HelpTopics
B116	content.HelpVotes	HelpArticles, Users
B117	content.LegalVersions	LegalDocuments, Users
B118	content.LegalAcceptances	LegalVersions, Users
B119	content.CookieConsents	Users
B120	content.NewsPosts	Users, Files
B121	notify.Notifications	Users
B122	notify.NotificationPreferences	Users

#	Table	Depends on
B123	`notify.EmailOutbox`	Users
B124	`notify.Broadcasts`	Users

Part C — Junction tables (many-to-many)

#	Table	Joins
C1	`auth.UserRoles`	Users ↔ Roles
C2	`auth.RolePermissions`	Roles ↔ Permissions
C3	`auth.UserPermissions`	Users ↔ Permissions (direct grants, e.g. Approvals to one admin)
C4	`ai.AgentPermissions`	Agents ↔ Permissions
C5	`pitch.PitchFacilities`	Pitches ↔ Facilities
C6	`pod.DesignColors`	Designs ↔ CatalogColors
C7	`pod.SellerFollowers`	Sellers ↔ Users
C8	`shop.WishlistItems`	Users ↔ Products / Designs
C9	`money.PromoCodePitches`	PromoCodes ↔ Pitches
C10	`fantasy.SeasonClubs`	Seasons ↔ Clubs
C11	`fantasy.GroupMembers`	FantasyGroups ↔ FantasyTeams
C12	`community.CommunityMembers`	Communities ↔ Users
C13	`community.PollVotes`	PollOptions ↔ Users

Totals: 25 lookup/independent tables, 124 child tables, 13 junction tables — **162 tables** in 16 schemas.

3. Part A — Lookup and independent tables

Columns are listed **in addition to** the tags. "Rules" uses: **PK**, **FK → table (ON DELETE)**, **UQ** unique, **DF** default, **CK** check.

A1 geo.Governorates — [L]

Column	Type	Null	Rules
Code	varchar(10)	NO	UQ (e.g. `CAI`, `GIZ`, `ALX`)
NameEn	nvarchar(80)	NO	UQ
NameAr	nvarchar(80)	NO	UQ

A2 auth.Permissions — [L]

Column	Type	Null	Rules
Code	varchar(80)	NO	UQ, CK `Code LIKE '%.%'` (e.g. `payouts.send`, `approvals.partners`)
Area	varchar(40)	NO	e.g. `approvals`, `money`, `content`, `community`, `fantasy`, `settings`, `audit`, `ai`
NameEn / NameAr	nvarchar(120)	NO	
DescriptionEn	nvarchar(400)	YES	
IsSensitive	bit	NO	DF 0 — money or personal data; shown in red in the CEO matrix

Index: `IX_Permissions_Area`.

A3 auth.Roles (Identity `AspNetRoles`)

Column	Type	Null	Rules
Id	uniqueidentifier	NO	PK, DF `NEWSEQUENTIALID()`
Name / NormalizedName	nvarchar(256)	NO	UQ on NormalizedName
ConcurrencyStamp	nvarchar(max)	YES	Identity
Kind	tinyint	NO	CK 0–2 (0 System, 1 Admin role made by the CEO, 2 Agent role)
DescriptionEn	nvarchar(400)	YES	
IsLocked	bit	NO	DF 0 — system roles (CEO, Customer…) cannot be deleted

A4 auth.Users (Identity `AspNetUsers`)

Column	Type	Null	Rules
Id	uniqueidentifier	NO	PK, DF `NEWSEQUENTIALID()`

Column	Type	Null	Rules
UserName / NormalizedUserName	nvarchar(256)	NO	UQ on NormalizedUserName
Email / NormalizedEmail	nvarchar(256)	NO	UQ on NormalizedEmail (filtered `WHERE IsDeleted = 0`)
EmailConfirmed, PhoneNumberConfirmed, TwoFactorEnabled, LockoutEnabled	bit	NO	Identity, DF 0 (LockoutEnabled DF 1)
PasswordHash, SecurityStamp, ConcurrencyStamp	nvarchar(max)	YES	Identity
PhoneNumber	varchar(20)	YES	UQ filtered `WHERE PhoneNumber IS NOT NULL`, CK `PhoneNumber LIKE '+20%'`
LockoutEnd	datetimeoffset	YES	Identity
AccessFailedCount	int	NO	DF 0
FullName	nvarchar(120)	NO	CK `LEN(FullName) >= 2`
AvatarFileId	bigint	YES	(FK added after `ops.Files` exists — NO ACTION)
Language	char(2)	NO	DF `'ar'`, CK IN (`'ar'`, `'en'`)
AccountType	tinyint	NO	DF 0, CK 0–2 (0 Person, 1 AI agent service account, 2 System)
Status	tinyint	NO	DF 0, CK 0–2 (0 Active, 1 Suspended, 2 Blocked from booking/ordering)
SuspendedReason	nvarchar(400)	YES	
MustPrepay	tinyint	NO	DF 0, CK 0–2 (0 Automatic, 1 Always, 2 Never) — COD rule for printed items
CodRefusals	int	NO	DF 0, CK ≥ 0
AdminNote	nvarchar(1000)	YES	private note by the Super App team
LastSeenAt	datetime2(0)	YES	
CreatedAt / UpdatedAt	datetime2(0)	NO / YES	DF SYSUTCDATETIME()
[S]			soft delete (data-protection deletion keeps money history anonymised)

Indexes: `IX_Users_Status`, `IX_Users_CreatedAt`.

A5 ops.Departments — [L]

Column	Type	Null	Rules
Key	varchar(30)	NO	UQ (`ops`, `support`, `bizdev`, `finance`, `legal`, `marketing`, `tech`, `aiops`)

Column	Type	Null	Rules
NameEn / NameAr	nvarchar(80)	NO	
Color	char(7)	NO	DF `'#0B6E4F'`, CK `Color LIKE '#%'`

A6 ops.Settings

Column	Type	Null	Rules
Key	varchar(80)	NO	**PK** (e.g. `commission.bookings`, `vat.rate`, `pod.holdDays`, `limits.payoutCeo`, `package.maxDiscount`, `loyalty.pointValue`)
Value	nvarchar(max)	NO	JSON, CK `ISJSON(Value) = 1`
Group	varchar(40)	NO	`money`, `limits`, `pod`, `loyalty`, `delivery`, `fantasy`, `popups` …
UpdatedAt	datetime2(0)	NO	DF SYSUTCDATETIME()
UpdatedById	uniqueidentifier	YES	no FK (the audit log records the change)

A7 ops.FeatureFlags — [B]

Column	Type	Null	Rules
Key	varchar(80)	NO	UQ — seeded kill switches: `kill.bookings`, `kill.shop`, `kill.pod`, `kill.payouts`, `kill.applications`, `kill.fantasy`, `kill.community`, `kill.games`, `kill.agents`; API groups such as `api.fantasy`, `api.agents`
Kind	tinyint	NO	CK 0–1 (0 Kill switch, 1 API group)
IsEnabled	bit	NO	DF 1
Reason	nvarchar(400)	YES	why it was switched off
ChangedById	uniqueidentifier	YES	

A8 ops.Integrations — `Id int IDENTITY` PK + CreatedAt/UpdatedAt as [B]

Column	Type	Null	Rules
Key	varchar(60)	NO	UQ (`payments.paymob`, `courier.bosta`, `email.smtp`, `football.apifootball`, `ai.claude`, `automation.n8n`, `storage.local`, `einvoice.eta`)
Category	tinyint	NO	CK 0–11 (Payment, Courier, Email, SMS, WhatsApp, Football data, AI model, Automation, Storage/E-invoice, **Geocoding**, **Weather**, **Holidays**) — keys such as `geo.nominatim`, `weather.openmeteo`, `holidays.nagerdate`
Name	nvarchar(80)	NO	
Mode	tinyint	NO	DF 0, CK 0–2 (0 Simulated, 1 Test/sandbox, 2 Live)
IsEnabled	bit	NO	DF 0

Column	Type	Null	Rules
IsPrimary	bit	NO	DF 0 — the active provider of its category (filtered UQ on `Category WHERE IsPrimary = 1`)
ConfigJson	nvarchar(max)	YES	non-secret settings, CK ISJSON
SecretName	varchar(120)	YES	name of the environment variable / encrypted secret — **the secret itself is never stored here**
SecretSetAt	datetime2(0)	YES	shown as "set / not set"
Status	tinyint	NO	DF 0, CK 0–3 (Unknown, Healthy, Degraded, Down)
LastCallAt / LastErrorAt	datetime2(0)	YES	
LastError	nvarchar(1000)	YES	
DailyQuota	int	YES	e.g. 100 for API-FOOTBALL free plan

A9 ops.WebhookSubscriptions — `Id int IDENTITY` PK + CreatedAt/UpdatedAt as [B]

Column	Type	Null	Rules
Name	nvarchar(80)	NO	e.g. "n8n — operations"
Url	nvarchar(500)	NO	CK `Url LIKE 'https://%'` (http allowed only in development)
Events	nvarchar(1000)	NO	JSON array of event types, CK ISJSON
SecretHash	varbinary(64)	NO	HMAC key (encrypted)
IsEnabled	bit	NO	DF 1
FailureCount	int	NO	DF 0 — auto-pause after 50 failures in a row

A10 ops.OutboxEvents

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
EventId	uniqueidentifier	NO	UQ, DF NEWID() — idempotency for receivers
Type	varchar(80)	NO	e.g. `booking.created`
PayloadJson	nvarchar(max)	NO	CK ISJSON
OccurredAt	datetime2(3)	NO	DF SYSUTCDATETIME()
Status	tinyint	NO	DF 0, CK 0–3 (Pending, Delivered, Failed, Skipped)
Attempts	int	NO	DF 0
NextAttemptAt	datetime2(0)	YES	

Index: `IX_OutboxEvents_Status_NextAttemptAt` (the dispatcher reads pending events).

A11 pitch.Facilities — [L]

Column	Type	Null	Rules
Key	varchar(30)	NO	UQ (`lights` , `parking` , `showers` , `lockers` , `cafe` , `water` , `wc` , `prayer` , `seats` , `wifi` , `ac` , `firstaid` , `camera`)
NameEn / NameAr	nvarchar(60)	NO	
Icon	varchar(30)	NO	

A12 pitch.CancellationPolicies — [L]

Column	Type	Null	Rules
Key	varchar(20)	NO	UQ (`flexible` , `standard` , `moderate` , `strict`)
NameEn / NameAr	nvarchar(40)	NO	
FullRefundHours	smallint	NO	CK ≥ 0 (2 / 6 / 24 / 48)
HalfRefundHours	smallint	NO	CK ≥ 0 and ≤ FullRefundHours (0 / 0 / 6 / 24)
SummaryEn / SummaryAr	nvarchar(300)	NO	

A13 shop.ProductCategories — [L]

Column	Type	Null	Rules
Key	varchar(40)	NO	UQ (`boots` , `kits` , `balls` …)
NameEn / NameAr	nvarchar(60)	NO	
Icon	varchar(30)	NO	
IsHidden	bit	NO	DF 0 — hides its products from the shop

A14 pod.DesignCategories — [L]

Column	Type	Null	Rules
Key	varchar(40)	NO	UQ
NameEn / NameAr	nvarchar(60)	NO	UQ on NameEn

A15 pod.SellerLevels — [L]

Column	Type	Null	Rules
Key	varchar(20)	NO	UQ (`starter` , `rising` , `pro` , `elite`)
NameEn / NameAr	nvarchar(40)	NO	
Rank	tinyint	NO	UQ, CK 0–9
SalesNeeded	int	NO	CK ≥ 0

Column	Type	Null	Rules
MinRating	decimal(2,1)	NO	DF 0, CK 0–5
ActiveDesigns	int	YES	NULL = unlimited, CK > 0
ActiveCodes	int	YES	NULL = unlimited
BaseDiscount	decimal(12,2)	NO	DF 0, CK ≥ 0
StoreCommission	decimal(5,2)	NO	CK 0–100
ReviewLane	tinyint	NO	CK 0–2 (Normal, Priority, Instant + spot check)
SearchBoost	tinyint	NO	DF 0, CK 0–5
SampleDiscount	decimal(5,2)	NO	DF 0, CK 0–100
FreeSamplesPerMonth	tinyint	NO	DF 0
FantasyDiscount	decimal(5,2)	NO	DF 0, CK 0–100
BulkUpload, TeamKits, EarlyAccess	bit	NO	DF 0
AnalyticsTier	tinyint	NO	CK 0–2 (Basic, Advanced, Advanced + export)
CalendarTier	tinyint	NO	CK 0–3
SupportTier	tinyint	NO	CK 0–2 (Normal, Priority, Account manager)

A16 pod.CatalogProducts — `Id int IDENTITY` PK + CreatedAt/UpdatedAt as [B], [S]

Column	Type	Null	Rules
Key	varchar(40)	NO	UQ (`pod-tee` , `pod-oversized` , `pod-hoodie`)
NameEn / NameAr	nvarchar(80)	NO	
Kind	tinyint	NO	CK 0–1 (0 Tee, 1 Hoodie)
DescriptionEn / DescriptionAr	nvarchar(400)	NO	
BasePrice	decimal(12,2)	NO	CK > 0
FactoryCost	decimal(12,2)	NO	CK > 0 and < BasePrice (the difference is Super App's margin)
BackPrintExtra	decimal(12,2)	NO	DF 0, CK ≥ 0
BigSizeExtra	decimal(12,2)	NO	DF 0, CK ≥ 0
MaxPrice	decimal(12,2)	NO	CK > BasePrice
HasBackPrint	bit	NO	DF 1
Model3dUrl	nvarchar(300)	YES	e.g. `models/tee.glb`
Status	tinyint	NO	DF 0, CK 0–2 (Active, Paused, Retired)

A17 fantasy.Competitions — [L]

Column	Type	Null	Rules
Key	varchar(10)	NO	UQ (`EGY` , `EPL`)
NameEn / NameAr	nvarchar(80)	NO	
ExternalLeagueId	int	NO	UQ — API-FOOTBALL league id
Country	nvarchar(40)	NO	
SquadBudget	decimal(6,1)	NO	DF 100.0, CK > 0
MaxPerClub	tinyint	NO	DF 3, CK 1–5

A18 fantasy.Clubs — [B]

Column	Type	Null	Rules
ExternalId	int	NO	UQ — API-FOOTBALL team id
NameEn / NameAr	nvarchar(80)	NO	
ShortName	varchar(5)	NO	
KitColor / KitColor2	char(7)	NO	DF `'#0D1411'` — shown instead of logos for now
LogoUrl	nvarchar(300)	YES	stays empty until logos are licensed

A19 fantasy.Sponsors — [B] [S]

Column	Type	Null	Rules
Name	nvarchar(100)	NO	UQ filtered
Tier	tinyint	NO	CK 0–2 (Gold, Silver, Bronze)
ContactEmail	nvarchar(256)	YES	
ContractValue	decimal(12,2)	NO	DF 0, CK ≥ 0
StartsOn / EndsOn	date	NO	CK EndsOn ≥ StartsOn
Color	char(7)	NO	

A20 content.LegalDocuments — [L]

Column	Type	Null	Rules
Key	varchar(30)	NO	UQ (`terms` , `privacy` , `refunds` , `waiver` , `seller` , `partner` , `cookies`)
TitleEn / TitleAr	nvarchar(120)	NO	
RequiresAcceptance	bit	NO	DF 1

A21 content.HelpTopics — [L]

Column	Type	Null	Rules
Key	varchar(30)	NO	UQ
NameEn / NameAr	nvarchar(80)	NO	
Icon	varchar(30)	NO	

A22 content.Banners — [B]

Column	Type	Null	Rules
Title	nvarchar(70)	NO	CK LEN ≥ 5
Badge	nvarchar(22)	YES	
CtaText	nvarchar(22)	YES	
Link	nvarchar(300)	NO	CK `Link LIKE '#/%'` (inside Super App only)
ImageKey	nvarchar(300)	NO	library key or file URL
IsActive	bit	NO	DF 1
SortOrder	int	NO	DF 0
ShowFrom / ShowUntil	date	YES	CK ShowUntil ≥ ShowFrom

A23 content.PopupAds — [B]

Column	Type	Null	Rules
TitleEn / TitleAr	nvarchar(60)	NO	
TextEn / TextAr	nvarchar(140)	NO	
CtaEn / CtaAr	nvarchar(24)	NO	
Link	nvarchar(300)	NO	CK `Link LIKE '#/%'`
Badge	nvarchar(24)	YES	
Color	char(7)	NO	CK `Color LIKE '#%'`
Shape	tinyint	NO	DF 0, CK 0–2 (Cube, Coin, 3D T-shirt)
Icons	varchar(200)	YES	comma-separated icon keys for the cube
HideFromSellers	bit	NO	DF 0
IsActive	bit	NO	DF 1
SortOrder	int	NO	DF 0
ShownCount / ClickedCount	int	NO	DF 0, CK ≥ 0

A24 community.BannedWords — [B]

Column	Type	Null	Rules
Word	nvarchar(60)	NO	UQ
Language	char(2)	NO	CK IN (`'ar'` , `'en'`)

A25 ops.PublicHolidays — `Id int IDENTITY` PK + CreatedAt/UpdatedAt as [B]

Column	Type	Null	Rules
Country	char(2)	NO	DF `'EG'`
Date	date	NO	UQ (Country, Date, NameEn)
NameEn / NameAr	nvarchar(120)	NO	
Kind	tinyint	NO	CK 0–2 (National, Religious, Observance)
IsDayOff	bit	NO	DF 1
IsEstimated	bit	NO	DF 0 — Islamic holidays waiting for the moon sighting
ConfirmedById	uniqueidentifier	YES	no FK (lookup table rule); the audit log records the confirmation
ConfirmedAt	datetime2(0)	YES	
Source	varchar(20)	NO	CK IN (`'nagerdate'` , `'calendarific'` , `'admin'`)

Index: `IX_PublicHolidays_Date` .

4. Part B — Child tables: accounts, operations, AI, pitches

B1 geo.Areas — [L]

Column	Type	Null	Rules
GovernorateId	int	NO	FK → geo.Governorates (NO ACTION)
NameEn / NameAr	nvarchar(80)	NO	UQ (GovernorateId, NameEn)
Lat / Lng	decimal(9,6)	YES	area centre for "near me"

Index: `IX_Areas_GovernorateId`.

B2–B5 Identity child tables

Standard ASP.NET Core Identity tables, renamed into the `auth` schema: **RoleClaims** (RoleId FK → Roles CASCADE), **UserClaims**, **UserLogins**, **UserTokens** (UserId FK → Users CASCADE). Columns as generated by Identity.

B6 auth.StaffLimits — [B]

Money limits per admin, CFO or agent's service account (above the limit an approval is created). | Column | Type | Null | Rules | |---|---|---|---| | UserId | uniqueidentifier | NO | UQ, FK → auth.Users (CASCADE) | | RefundLimit | decimal(12,2) | NO | DF 0, CK ≥ 0 | | CreditLimit | decimal(12,2) | NO | DF 0, CK ≥ 0 | | PayoutLimit | decimal(12,2) | NO | DF 0, CK ≥ 0 | | CanSecondApprove | bit | NO | DF 0 — the CFO's second approval |

B7 auth.UserAddresses — [B]

Column	Type	Null	Rules
UserId	uniqueidentifier	NO	FK → auth.Users (CASCADE)
AreaId	int	NO	FK → geo.Areas (NO ACTION)
Label	nvarchar(40)	NO	DF N'Home'
Street	nvarchar(200)	NO	
Building / Floor / Apartment	nvarchar(20)	YES	
Landmark	nvarchar(120)	YES	
Phone	varchar(20)	NO	CK `LIKE '+20%'`
IsDefault	bit	NO	DF 0 — filtered UQ (UserId) WHERE IsDefault = 1

B8 ops.Files — [B] [S]

Every uploaded or generated file (photos, artwork, ID scans, PDFs). The bytes live in object storage, never in the database. | Column | Type | Null | Rules | |---|---|---|---| | OwnerUserId | uniqueidentifier | YES | FK → auth.Users (NO ACTION) | | StorageKey | nvarchar(400) | NO | UQ — path in the storage container | | Container | varchar(40) | NO | `public`, `private`, `artwork`, `kyc`, `receipts`, `labels` | | FileName | nvarchar(200) | NO | | | ContentType | varchar(100) | NO | CK in an allowed list (image/png, image/jpeg, image/webp, image/svg+xml, application/pdf, model/gltf-binary) | | SizeBytes | bigint | NO | CK > 0 and ≤ 20 MB | | Width / Height | int | YES | images | | Sha256 | char(64) | NO | duplicate and tamper check | | IsPrivate | bit | NO | DF 1 — private files are served only through signed, time-limited URLs |

After this table exists, `auth.Users.AvatarFileId` gets its FK (NO ACTION).

B9 auth.KycDocuments — [B]

Column	Type	Null	Rules
UserId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
NationalId	char(14)	NO	CK `NationalId NOT LIKE '%[^0-9]%'` (14 digits); stored encrypted
FrontFileId / BackFileId	bigint	NO	FK → ops.Files (NO ACTION)
Status	tinyint	NO	DF 0, CK 0–2 (Pending, Verified, Rejected)
ReviewedById	uniqueidentifier	YES	FK → auth.Users (NO ACTION)
ReviewedAt	datetime2(0)	YES	
RejectReason	nvarchar(400)	YES	CK required when Status = 2

Index: `IX_KycDocuments_UserId_Status`.

B10 auth.Applications — [B]

Requests to become a pitch owner, a seller or a factory. | Column | Type | Null | Rules | |---|---|---|---| | Code | varchar(12) | NO | UQ | | ApplicantId | uniqueidentifier | NO | FK → auth.Users (NO ACTION) | | Type | tinyint | NO | CK 0–2 (PitchOwner, Seller, Factory) | | AnswersJson | nvarchar(max) | NO | the form answers (venue, store name, payout method…), CK ISJSON | | AgreementVersionId | bigint | NO | FK → content.LegalVersions (added after B117, NO ACTION) | | Status | tinyint | NO | DF 0, CK 0–3 (Pending, Approved, Rejected, Withdrawn) | | ReviewedById | uniqueidentifier | YES | FK → auth.Users (NO ACTION) | | ReviewedAt | datetime2(0) | YES | | | RejectReason | nvarchar(400) | YES | CK required when Status = 2 |

Indexes: `IX_Applications_Status_Type_CreatedAt`; filtered UQ (ApplicantId, Type) WHERE Status = 0 — one open application per type.

B11 auth.ApplicationFiles — [B]

Column	Type	Null	Rules
ApplicationId	bigint	NO	FK → auth.Applications (CASCADE)
FileId	bigint	NO	FK → ops.Files (NO ACTION)

Column	Type	Null	Rules
Purpose	tinyint	NO	CK 0–3 (ID front, ID back, Venue photo, Document)

B12 ops.OrgPositions — [B]

Boxes of the company pyramid. A human holds a position through `HolderUserId`; an AI agent through `ai.Agents.PositionId`. | Column | Type | Null | Rules | |---|---|---|---| | DepartmentId | int | YES | FK → ops.Departments (NO ACTION); NULL for the CEO | | ReportsToId | bigint | YES | FK → ops.OrgPositions (NO ACTION, self) | | TitleEn / TitleAr | nvarchar(120) | NO | | | HolderUserId | uniqueidentifier | YES | FK → auth.Users (SET NULL) | | DutiesJson | nvarchar(max) | YES | list of duties, CK ISJSON | | SortOrder | int | NO | DF 0 |

Index: `IX_OrgPositions_ReportsToId`.

B13 ai.Agents — [B] [S]

Column	Type	Null	Rules
ServiceUserId	uniqueidentifier	NO	UQ, FK → auth.Users (NO ACTION) — the agent's service account
PositionId	bigint	YES	UQ filtered, FK → ops.OrgPositions (SET NULL)
Name	nvarchar(60)	NO	UQ filtered (e.g. "Atlas")
TitleEn / TitleAr	nvarchar(120)	NO	
Description	nvarchar(2000)	NO	
Model	varchar(60)	NO	e.g. `claude-sonnet-5-5`
Platform	tinyint	NO	DF 0, CK 0–2 (n8n, Direct API, MCP)
MoneyLimit	decimal(12,2)	NO	DF 0, CK ≥ 0
MonthlyBudget	decimal(12,2)	NO	DF 0, CK ≥ 0 (AI cost)
Status	tinyint	NO	DF 1, CK 0–2 (Active, Paused, Stopped by emergency switch)
PausedReason	nvarchar(400)	YES	

B14 ai.ApiKeys — [B]

Column	Type	Null	Rules
AgentId	bigint	YES	FK → ai.Agents (CASCADE); NULL for an n8n system key
Name	nvarchar(80)	NO	
Prefix	char(8)	NO	UQ — shown in the UI (`sa_ab12…`)
KeyHash	varbinary(64)	NO	SHA-256 of the key — the key is shown once and never stored
Scopes	nvarchar(1000)	YES	optional extra narrowing, JSON
ExpiresAt	datetime2(0)	YES	
RevokedAt	datetime2(0)	YES	

Column	Type	Null	Rules
LastUsedAt	datetime2(0)	YES	

B15 ai.AgentTasks — [B]

Column	Type	Null	Rules
AgentId	bigint	NO	FK → ai.Agents (CASCADE)
Title	nvarchar(120)	NO	
Description	nvarchar(2000)	NO	
Trigger	tinyint	NO	CK 0–2 (Event, Schedule, Manual)
EventType	varchar(80)	YES	CK required when Trigger = 0
Cron	varchar(60)	YES	CK required when Trigger = 1
ExpectedOutput	nvarchar(1000)	YES	
IsEnabled	bit	NO	DF 1
LastRunAt	datetime2(0)	YES	

B16 ai.AgentRuns

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
AgentId	bigint	NO	FK → ai.Agents (NO ACTION)
TaskId	bigint	YES	FK → ai.AgentTasks (SET NULL)
StartedAt / FinishedAt	datetime2(0)	NO / YES	DF SYSUTCDATETIME()
Result	tinyint	NO	DF 0, CK 0–4 (Running, Done, Waiting for approval, Failed, Cancelled)
Summary	nvarchar(2000)	YES	
InputTokens / OutputTokens	int	NO	DF 0, CK ≥ 0
CostUsd	decimal(10,4)	NO	DF 0, CK ≥ 0

Index: `IX_AgentRuns_AgentId_StartedAt`.

B17 ops.AuditLogs (append-only)

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK

Column	Type	Null	Rules
At	datetime2(3)	NO	DF SYSUTCDATETIME()
ActorUserId	uniqueidentifier	YES	FK → auth.Users (NO ACTION)
ActorAgentId	bigint	YES	FK → ai.Agents (NO ACTION)
System	tinyint	NO	CK 0–6 (Site, Partner, Seller, Factory, Admin, CEO, API/Agent)
Action	varchar(80)	NO	e.g. `payout.sent` , `pitch.approved`
EntityType	varchar(60)	YES	
EntityId	varchar(40)	YES	
Summary	nvarchar(500)	NO	human-readable line
DataJson	nvarchar(max)	YES	before/after values, CK ISJSON
Ip	varchar(45)	YES	
CorrelationId	varchar(64)	YES	

Indexes: `IX_AuditLogs_At` , `IX_AuditLogs_EntityType_EntityId` , `IX_AuditLogs_ActorUserId_At` . The application database user has **no UPDATE or DELETE** permission on this table.

B18 ops.IntegrationUsage

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
IntegrationId	int	NO	FK → ops.Integrations (CASCADE)
Day	date	NO	UQ (IntegrationId, Day)
Calls / Errors	int	NO	DF 0, CK ≥ 0
QuotaUsed / QuotaLimit	int	YES	from the provider's headers (API-FOOTBALL usage bar)

B19 ops.WebhookDeliveries

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
SubscriptionId	int	NO	FK → ops.WebhookSubscriptions (CASCADE)
OutboxEventId	bigint	NO	FK → ops.OutboxEvents (NO ACTION)
Attempt	tinyint	NO	CK ≥ 1
StatusCode	smallint	YES	
DurationMs	int	YES	
Error	nvarchar(1000)	YES	
At	datetime2(0)	NO	DF SYSUTCDATETIME()

B20 ops.Goals — [B] · B21 ops.KeyResults — [B]

Table	Column	Type	Null	Rules
Goals	DepartmentId	int	YES	FK → ops.Departments (NO ACTION)
Goals	OwnerUserId	uniqueidentifier	YES	FK → auth.Users (SET NULL)
Goals	Title	nvarchar(200)	NO	
Goals	Quarter	char(7)	NO	e.g. `2026-Q4` , CK `LIKE '[0-9][0-9][0-9][0-9]-Q[1-4]'`
KeyResults	GoalId	bigint	NO	FK → ops.Goals (CASCADE)
KeyResults	Title	nvarchar(200)	NO	
KeyResults	Metric	varchar(60)	YES	live metric key, e.g. `bookings.count`
KeyResults	Target / Current	decimal(14,2)	NO	CK Target > 0

B22 ops.Budgets — [B]

Column	Type	Null	Rules
DepartmentId	int	NO	FK → ops.Departments (NO ACTION)
Month	date	NO	first day of the month; UQ (DepartmentId, Month)
Amount	decimal(12,2)	NO	CK ≥ 0

B23 ops.ComplianceItems — [B]

Column	Type	Null	Rules
DepartmentId	int	YES	FK → ops.Departments (NO ACTION)
OwnerUserId	uniqueidentifier	YES	FK → auth.Users (SET NULL)
Title	nvarchar(200)	NO	e.g. "VAT return", "Data-protection licence renewal"
DueOn	date	NO	
Repeat	tinyint	NO	CK 0–3 (Once, Monthly, Quarterly, Yearly)
DoneAt	datetime2(0)	YES	

B24 pitch.Partners — [B] [S]

Column	Type	Null	Rules
UserId	uniqueidentifier	NO	UQ, FK → auth.Users (NO ACTION)
BusinessName	nvarchar(120)	NO	
BookingCommission	decimal(5,2)	YES	per-partner deal; NULL = platform default; CK 0–100
StoreCommission	decimal(5,2)	YES	CK 0–100
PayoutMethod	tinyint	NO	CK 0–1 (Bank, InstaPay)

Column	Type	Null	Rules
PayoutDetail	nvarchar(200)	NO	stored encrypted
Status	tinyint	NO	DF 0, CK 0–1 (Active, Suspended)
SuspendedReason	nvarchar(400)	YES	
SlotPostingBlocked	bit	NO	DF 0

B25 pitch.PartnerStaff — [B]

Column	Type	Null	Rules
PartnerId	bigint	NO	FK → pitch.Partners (CASCADE)
UserId	uniqueidentifier	NO	FK → auth.Users (NO ACTION); UQ (PartnerId, UserId)
Role	tinyint	NO	CK 0–2 (Manager, Check-in staff, Accountant)
IsActive	bit	NO	DF 1

B26 pitch.Venues — [B] [S]

Column	Type	Null	Rules
PartnerId	bigint	NO	FK → pitch.Partners (NO ACTION)
Name	nvarchar(120)	NO	
AreaId	int	NO	FK → geo.Areas (NO ACTION)
Address	nvarchar(300)	NO	
Lat / Lng	decimal(9,6)	NO	CK Lat 22–32, Lng 24–37 (Egypt)
Phone	varchar(20)	YES	

Indexes: `IX_Venues_PartnerId`, `IX_Venues_AreaId`.

B27 pitch.Pitches — [B] [S] [V]

Column	Type	Null	Rules
VenueId	bigint	NO	FK → pitch.Venues (NO ACTION)
Slug	varchar(80)	NO	UQ filtered (used in links)
Name	nvarchar(120)	NO	
Format	tinyint	NO	CK IN (5, 7, 11)
Surface	tinyint	NO	CK 0–2 (Artificial turf, Natural grass, Indoor court)
IsIndoor	bit	NO	DF 0
PricePerHour	decimal(12,2)	NO	CK > 0
PeakPricePerHour	decimal(12,2)	YES	CK > 0

Column	Type	Null	Rules
PeakFromHour	tinyint	YES	CK 0–23
DescriptionEn / DescriptionAr	nvarchar(2000)	YES	
CancellationPolicyId	int	NO	FK → pitch.CancellationPolicies (NO ACTION)
AllowCash	bit	NO	DF 1
Status	tinyint	NO	DF 0, CK 0–4 (Waiting for review, Live, Rejected, Closed by owner, Closed by admin)
ReviewedById	uniqueidentifier	YES	FK → auth.Users (NO ACTION)
RejectReason	nvarchar(400)	YES	
Rating	decimal(3,2)	NO	DF 0, CK 0–5 (cached average)
ReviewsCount	int	NO	DF 0

Indexes: `IX_Pitches_Status_VenueId`, `IX_Pitches_Format_PricePerHour` (search).

B28 pitch.PitchPhotos — [B]

Column	Type	Null	Rules
PitchId	bigint	NO	FK → pitch.Pitches (CASCADE)
FileId	bigint	NO	FK → ops.Files (NO ACTION)
SortOrder	int	NO	DF 0

B29 pitch.OpeningHours

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
PitchId	bigint	NO	FK → pitch.Pitches (CASCADE)
DayOfWeek	tinyint	NO	CK 0–6 (0 = Sunday); UQ (PitchId, DayOfWeek)
OpenHour	tinyint	NO	CK 0–23
CloseHour	tinyint	NO	CK 1–24 and > OpenHour (24 = midnight; overnight is split into two rows)
IsClosed	bit	NO	DF 0

B30 pitch.Closures — [B] [A]

Column	Type	Null	Rules
PitchId	bigint	NO	FK → pitch.Pitches (CASCADE)
FromDate / ToDate	date	NO	CK ToDate ≥ FromDate
FromHour / ToHour	tinyint	YES	NULL = whole days; CK 0–24, ToHour > FromHour

Column	Type	Null	Rules
Reason	nvarchar(200)	NO	

Index: `IX_Closures_PitchId_FromDate`.

B31 pitch.PackageOptions — [B]

The package sizes a pitch offers, each with its own discount (approved by an admin). | Column | Type | Null | Rules | |---|---|---|---| | PitchId | bigint | NO | FK → pitch.Pitches (CASCADE) | | MatchCount | tinyint | NO | CK 1–12; UQ (PitchId, MatchCount) | | DiscountPercent | decimal(5,2) | NO | DF 0, CK 0–100 (≤ the CEO's maximum, checked in the service) | | Status | tinyint | NO | DF 0, CK 0–3 (Waiting for approval, Approved, Rejected, Disabled) | | ApprovedDiscount | decimal(5,2) | YES | the live value while a change waits for approval | | ReviewedById | uniqueidentifier | YES | FK → auth.Users (NO ACTION) | | ReviewedAt | datetime2(0) | YES | | | RejectReason | nvarchar(400) | YES | |

B32 pitch.Offers — [B] [A]

Quiet-hour discounts created from insights. | Column | Type | Null | Rules | |---|---|---|---| | PitchId | bigint | NO | FK → pitch.Pitches (CASCADE) | | DaysMask | tinyint | NO | bit mask of weekdays, CK 1–127 | | FromHour / ToHour | tinyint | NO | CK 0–24, ToHour > FromHour | | DiscountPercent | decimal(5,2) | NO | CK > 0 and ≤ 90 | | StartsOn / EndsOn | date | NO | CK EndsOn ≥ StartsOn | | IsActive | bit | NO | DF 1 |

5. Part B — Child tables: stores, sellers, catalogue, factory, designs, products

B33 shop.Stores — [B] [S]

Column	Type	Null	Rules
OwnerUserId	uniqueidentifier	YES	FK → auth.Users (NO ACTION); NULL = the official Super App Store
Type	tinyint	NO	CK 0–2 (Official, Pitch partner store, Seller store)
Handle	varchar(40)	NO	UQ filtered, CK `Handle NOT LIKE '%[^a-z0-9-]%'` — used in `/s/{handle}`
Name	nvarchar(80)	NO	
Tagline	nvarchar(120)	YES	
About	nvarchar(1000)	YES	
LogoFileId / CoverFileId	bigint	YES	FK → ops.Files (NO ACTION)
CoverPositionY	tinyint	NO	DF 50, CK 0–100
AccentColor	char(7)	NO	DF `'#17795A'`
Status	tinyint	NO	DF 0, CK 0–2 (Active, On break, Suspended)
BreakUntil	date	YES	
BreakNote	nvarchar(200)	YES	

Filtered UQ: only one row with `Type = 0` (official store).

B34 pod.Sellers — [B] [S] [V]

Column	Type	Null	Rules
UserId	uniqueidentifier	NO	UQ, FK → auth.Users (NO ACTION)
StoreId	bigint	NO	UQ, FK → shop.Stores (NO ACTION)
LevelId	int	NO	FK → pod.SellerLevels (NO ACTION)
Mode	tinyint	NO	CK 0–2 (Print on demand, Own products, Both)
Status	tinyint	NO	DF 0, CK 0–2 (Active, Suspended, Frozen)
FrozenUntil	datetime2(0)	YES	copyright strike freeze (90 days)
GraceUntil	date	YES	grace month at the level review
LifetimeSales	int	NO	DF 0, CK ≥ 0 (delivered items)
Rating	decimal(3,2)	NO	DF 0, CK 0–5
FeaturedDesignId	bigint	YES	FK → pod.Designs (added after B46, NO ACTION)

Column	Type	Null	Rules
FeaturedByAdmin	bit	NO	DF 0
Tags	nvarchar(300)	YES	
PayoutMethod	tinyint	NO	CK 0–1 (Bank, InstaPay)
PayoutDetail	nvarchar(200)	NO	encrypted

Index: `IX_Sellers_LevelId_Status`.

B35 pod.SellerStrikes — [B] [A]

Column	Type	Null	Rules
SellerId	bigint	NO	FK → pod.Sellers (NO ACTION)
DesignId	bigint	YES	FK → pod.Designs (added after B46, SET NULL)
Reason	nvarchar(400)	NO	
ExpiresAt	datetime2(0)	NO	
RemovedAt	datetime2(0)	YES	
RemovedReason	nvarchar(400)	YES	

B36 pod.LevelReviews — [B]

Column	Type	Null	Rules
SellerId	bigint	NO	FK → pod.Sellers (CASCADE)
Month	date	NO	UQ (SellerId, Month)
FromLevelId / ToLevelId	int	NO	FK → pod.SellerLevels (NO ACTION)
Action	tinyint	NO	CK 0–3 (Kept, Promoted, Grace month, Lowered)
Note	nvarchar(400)	YES	

B37 pod.CatalogColors — [B]

Column	Type	Null	Rules
CatalogProductId	int	NO	FK → pod.CatalogProducts (CASCADE)
NameEn / NameAr	nvarchar(40)	NO	UQ (CatalogProductId, NameEn)
Hex	char(7)	NO	CK `Hex LIKE '#%'`
IsActive	bit	NO	DF 1 (the factory can pause a colour)
SortOrder	int	NO	DF 0

B38 pod.CatalogSizes — [B]

Sizes with the size chart (cm, measured flat). | Column | Type | Null | Rules | |---|---|---|---| | CatalogProductId | int | NO | FK → pod.CatalogProducts (CASCADE) | | Size | varchar(5) | NO | UQ (CatalogProductId, Size) — S, M, L, XL, 2XL, 3XL | | IsBigSize | bit | NO | DF 0 — costs `BigSizeExtra` | | ChestCm / LengthCm / SleeveCm | decimal(4,1) | NO | CK > 0 | | SortOrder | int | NO | DF 0 |

B39 pod.CatalogDetails — [B]

Column	Type	Null	Rules
CatalogProductId	int	NO	FK → pod.CatalogProducts (CASCADE)
LabelEn / LabelAr	nvarchar(40)	NO	Fabric, Fit, Print, Neck, Care, Details
ValueEn / ValueAr	nvarchar(300)	NO	
SortOrder	int	NO	DF 0

B40 pod.CatalogPrintAreas — [B]

Column	Type	Null	Rules
CatalogProductId	int	NO	FK → pod.CatalogProducts (CASCADE)
Side	tinyint	NO	CK 0–1 (Front, Back); UQ (CatalogProductId, Side)
X / Y / Width / Height	decimal(5,4)	NO	fractions of the 400×440 template, CK 0–1
PrintWidthCm	decimal(4,1)	NO	DF 30, CK > 0

B41 factory.Factories — [B] [S]

Column	Type	Null	Rules
OwnerUserId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
Name	nvarchar(120)	NO	UQ filtered
Address	nvarchar(300)	NO	
AreaId	int	NO	FK → geo.Areas (NO ACTION)
CapacityPerDay	int	NO	CK > 0
SlaHours	smallint	NO	DF 48, CK 1–240
Techniques	varchar(60)	NO	DF `'DTF'`
IntakePaused	bit	NO	DF 0
Status	tinyint	NO	DF 0, CK 0–2 (Active, Paused by admin, Suspended)
PayoutDetail	nvarchar(200)	NO	encrypted

B42 factory.FactoryStaff — [B] [S]

Column	Type	Null	Rules
FactoryId	bigint	NO	FK → factory.Factories (NO ACTION)
Name	nvarchar(80)	NO	UQ (FactoryId, Name) filtered
Role	tinyint	NO	CK 0–4 (Owner, Supervisor, Printer, Quality check, Packer)
PinHash	varbinary(64)	NO	hashed 4-digit PIN
FailedPins	tinyint	NO	DF 0
LockedUntil	datetime2(0)	YES	
IsActive	bit	NO	DF 1

B43 factory.FactoryDevices — [B]

Column	Type	Null	Rules
FactoryId	bigint	NO	FK → factory.Factories (CASCADE)
Name	nvarchar(60)	NO	e.g. "Packing tablet"
TokenHash	varbinary(64)	NO	UQ — device registration token
LastSeenAt	datetime2(0)	YES	
RevokedAt	datetime2(0)	YES	

B44 factory.FactoryPrices — [B]

Column	Type	Null	Rules
FactoryId	bigint	NO	FK → factory.Factories (CASCADE)
CatalogProductId	int	NO	FK → pod.CatalogProducts (NO ACTION); UQ (FactoryId, CatalogProductId)
PricePerPiece	decimal(12,2)	NO	CK > 0
BackPrintPrice	decimal(12,2)	NO	DF 0, CK ≥ 0

B45 pod.CatalogChangeRequests — [B]

Factory changes to the catalogue wait for an admin; only approved changes reach sellers. | Column | Type | Null | Rules | |---|---|---|---| | CatalogProductId | int | NO | FK → pod.CatalogProducts (NO ACTION) | | FactoryId | bigint | NO | FK → factory.Factories (NO ACTION) | | RequestedByStaffId | bigint | YES | FK → factory.FactoryStaff (NO ACTION) | | Kind | tinyint | NO | CK 0–4 (Price, Colour, Size, Pause product, Details) | | ChangeJson | nvarchar(max) | NO | old and new values, CK ISJSON | | Reason | nvarchar(400) | NO | | | Status | tinyint | NO | DF 0, CK 0–2 (Pending, Approved, Rejected) | | ReviewedById | uniqueidentifier | YES | FK → auth.Users (NO ACTION) | | ReviewedAt | datetime2(0) | YES | | | ReviewNote | nvarchar(400) | YES | |

Index: `IX_CatalogChangeRequests_Status`.

B46 pod.Designs — [B] [S] [V]

Column	Type	Null	Rules
SellerId	bigint	NO	FK → pod.Sellers (NO ACTION)
CatalogProductId	int	NO	FK → pod.CatalogProducts (NO ACTION)
CategoryId	int	YES	FK → pod.DesignCategories (NO ACTION); CK required to submit (service)
Title	nvarchar(60)	NO	DF N''
Description	nvarchar(500)	YES	
Price	decimal(12,2)	NO	CK ≥ 0 (≥ base and ≤ max checked in the service, because the base depends on the seller's level)
FrontArtFileId / BackArtFileId	bigint	YES	FK → ops.Files (NO ACTION); CK at least one is set to submit
FrontX / FrontY / FrontW	decimal(5,4)	YES	placement on the front, CK 0–1, all or none
BackX / BackY / BackW	decimal(5,4)	YES	placement on the back, CK 0–1, all or none
Status	tinyint	NO	DF 0, CK 0–5 (Draft, In review, Live, Rejected, Paused, Removed)
RejectReason	nvarchar(400)	YES	
SubmittedAt / ReviewedAt	datetime2(0)	YES	
ReviewedById	uniqueidentifier	YES	FK → auth.Users (NO ACTION)
SpotCheck	bit	NO	DF 0 (Elite instant publish)
Views / Sales	int	NO	DF 0, CK ≥ 0

Indexes: `IX_Designs_Status_CategoryId`, `IX_Designs_SellerId_Status`, `IX_Designs_SubmittedAt` (review queue).

B47 pod.DesignTags

Column	Type	Null	Rules
DesignId	bigint	NO	PK part, FK → pod.Designs (CASCADE)
Tag	nvarchar(40)	NO	PK part (DesignId, Tag)

Index: `IX_DesignTags_Tag` (tag search).

B48 pod.DesignReports — [B]

Column	Type	Null	Rules
DesignId	bigint	NO	FK → pod.Designs (NO ACTION)

Column	Type	Null	Rules
ReporterUserId	uniqueidentifier	YES	FK → auth.Users (SET NULL)
Reason	tinyint	NO	CK 0–3 (Copyright, Trademark, Offensive, Other)
Detail	nvarchar(600)	YES	
Contact	nvarchar(120)	YES	
Status	tinyint	NO	DF 0, CK 0–2 (Open, Taken down, Dismissed)
HandledById	uniqueidentifier	YES	FK → auth.Users (NO ACTION)
HandledAt	datetime2(0)	YES	

B49 shop.Products — [B] [S] [A]

Column	Type	Null	Rules
StoreId	bigint	NO	FK → shop.Stores (NO ACTION)
CategoryId	int	NO	FK → shop.ProductCategories (NO ACTION)
Slug	varchar(120)	NO	UQ filtered
Name	nvarchar(150)	NO	
Brand	nvarchar(80)	YES	
Description	nvarchar(2000)	YES	
Price	decimal(12,2)	NO	CK > 0
CompareAtPrice	decimal(12,2)	YES	CK > Price
LowStockAt	int	NO	DF 5, CK ≥ 0
Status	tinyint	NO	DF 0, CK 0–2 (Draft, Published, Unpublished)
HiddenByAdmin	bit	NO	DF 0
HiddenReason	nvarchar(400)	YES	CK required when HiddenByAdmin = 1
Rating	decimal(3,2)	NO	DF 0

Indexes: `IX_Products_StoreId_Status`, `IX_Products_CategoryId_Status_Price`.

B50 shop.ProductImages — [B]

Column	Type	Null	Rules
ProductId	bigint	NO	FK → shop.Products (CASCADE)
FileId	bigint	NO	FK → ops.Files (NO ACTION)
SortOrder	int	NO	DF 0

B51 shop.ProductVariants — [B] [V]

Column	Type	Null	Rules
ProductId	bigint	NO	FK → shop.Products (CASCADE)
Size	nvarchar(20)	NO	DF N'One size'
ColorName	nvarchar(40)	NO	DF N'Standard'
ColorHex	char(7)	YES	
Sku	varchar(40)	NO	UQ
Stock	int	NO	DF 0, CK ≥ 0
Reserved	int	NO	DF 0, CK ≥ 0 and ≤ Stock (held during checkout)
PriceOverride	decimal(12,2)	YES	CK > 0

UQ (ProductId, Size, ColorName).

B52 shop.ProductSpecs

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
ProductId	bigint	NO	FK → shop.Products (CASCADE)
Label	nvarchar(60)	NO	
Value	nvarchar(200)	NO	
SortOrder	int	NO	DF 0

6. Part B — Child tables: money, bookings, orders, factory jobs, shipping

B53 money.Wallets — [B] [V]

Column	Type	Null	Rules
OwnerUserId	uniqueidentifier	YES	FK → auth.Users (NO ACTION); NULL for the platform wallets
Type	tinyint	NO	CK 0–6 (Customer, Partner bookings, Partner store, Seller, Factory, Platform revenue, Platform clearing)
Currency	char(3)	NO	DF `'EGP'`
PendingBalance / AvailableBalance / SettledTotal	decimal(14,2)	NO	DF 0 — **cached** from the ledger (rebuilt nightly and checked)

UQ (OwnerUserId, Type). The balances may be negative only for platform wallets (CK `AvailableBalance >= 0 OR Type >= 5`).

B54 money.PromoCodes — [B] [S] [A]

Column	Type	Null	Rules
Code	varchar(30)	NO	UQ filtered, CK `Code NOT LIKE '%[^A-Z0-9]%'`
Funding	tinyint	NO	CK 0–1 (Owner-funded, Super App-funded)
OwnerUserId	uniqueidentifier	YES	FK → auth.Users (NO ACTION); CK required when Funding = 0
StoreId	bigint	YES	FK → shop.Stores (NO ACTION) — store codes
Scope	tinyint	NO	CK 0–2 (Bookings, Shop, Both)
DiscountType	tinyint	NO	CK 0–2 (Percent, Fixed, Free delivery)
Value	decimal(12,2)	NO	CK ≥ 0; CK ≤ 100 when Percent
MinSpend	decimal(12,2)	NO	DF 0, CK ≥ 0
MaxUses	int	YES	NULL = unlimited, CK > 0
PerCustomer	int	YES	CK > 0
UsesCount	int	NO	DF 0, CK ≥ 0
StartsOn / EndsOn	date	NO / YES	CK EndsOn ≥ StartsOn
Status	tinyint	NO	DF 0, CK 0–1 (Active, Paused)
Label / Note	nvarchar(120)	YES	

B55 money.Payments — [B] [V]

Column	Type	Null	Rules
Code	varchar(16)	NO	UQ — shown on receipts
PayerUserId	uniqueidentifier	YES	FK → auth.Users (NO ACTION)
Provider	varchar(40)	NO	integration key, e.g. `payments.simulated`, `payments.paymob`
ProviderRef	varchar(100)	YES	UQ filtered — the gateway transaction id
IdempotencyKey	varchar(64)	NO	UQ — retries never charge twice
Purpose	tinyint	NO	CK 0–7 (Booking, Package, Game share, Tournament fee, Shop order, Fantasy Pass, Sample, Wallet top-up)
Method	tinyint	NO	CK 0–5 (Card, Wallet app, InstaPay, Fawry cash, Cash at venue/COD, Super App wallet)
Amount	decimal(12,2)	NO	CK > 0
VatAmount	decimal(12,2)	NO	DF 0, CK ≥ 0 and ≤ Amount
Currency	char(3)	NO	DF `'EGP'`
Status	tinyint	NO	DF 0, CK 0–5 (Created, Pending, Paid, Failed, Refunded, Partly refunded)
PaidAt	datetime2(0)	YES	set **only** from the verified webhook
RawWebhookJson	nvarchar(max)	YES	CK ISJSON — kept for disputes

Index: `IX_Payments_PayerUserId_CreatedAt`, `IX_Payments_Status`.

B56 booking.Packages — [B] [V]

One QR for 1–12 weekly matches. | Column | Type | Null | Rules | |---|---|---|---| | Code | varchar(12) | NO | UQ | | PitchId | bigint | NO | FK → pitch.Pitches (NO ACTION) | | CustomerId | uniqueidentifier | NO | FK → auth.Users (NO ACTION) | | PackageOptionId | bigint | NO | FK → pitch.PackageOptions (NO ACTION) | | PaymentId | bigint | YES | FK → money.Payments (NO ACTION) | | MatchCount | tinyint | NO | CK 1–12 (copied from the option at purchase) | | UsedCount | tinyint | NO | DF 0, CK `UsedCount <= MatchCount` | | DayOfWeek | tinyint | NO | CK 0–6 | | StartHour / Duration | tinyint | NO | CK StartHour 0–23, Duration 1–4 | | StartsOn / EndsOn | date | NO | CK EndsOn ≥ StartsOn | | DiscountPercent | decimal(5,2) | NO | price snapshot, CK 0–100 | | TotalPrice | decimal(12,2) | NO | CK > 0 | | QrSecretHash | varbinary(64) | NO | signature key part of the QR token | | ExceptionUsed | bit | NO | DF 0 — one admin exception per package | | ExceptionById | uniqueidentifier | YES | FK → auth.Users (NO ACTION) | | Status | tinyint | NO | DF 0, CK 0–3 (Active, Finished, Expired, Cancelled) |

B57 booking.Series — [B]

Venue memberships (team plays every week, renews automatically). | Column | Type | Null | Rules | |---|---|---|---| | PitchId | bigint | NO | FK → pitch.Pitches (NO ACTION) | | TeamName | nvarchar(80) | NO | | | ContactUserId | uniqueidentifier | YES | FK → auth.Users (SET NULL) | | ContactPhone | varchar(20) | YES | | | DayOfWeek / StartHour / Duration | tinyint | NO | CK 0–6 / 0–23 / 1–4 | | AutoRenew | bit | NO | DF 1 | | Status | tinyint | NO | DF 0, CK 0–1 (Active, Ended) |

B58 booking.Bookings — [B] [V]

Column	Type	Null	Rules
Code	varchar(12)	NO	UQ (e.g. `SA-5P542931`)
PitchId	bigint	NO	FK → pitch.Pitches (NO ACTION)
CustomerId	uniqueidentifier	YES	FK → auth.Users (NO ACTION); NULL for walk-ins
WalkInName / WalkInPhone	nvarchar(80) / varchar(20)	YES	
Source	tinyint	NO	CK 0–5 (App, Walk-in, Package, Series, Tournament, Free slot)
PackageId	bigint	YES	FK → booking.Packages (NO ACTION)
SeriesId	bigint	YES	FK → booking.Series (NO ACTION)
Date	date	NO	
StartHour	tinyint	NO	CK 0–23
Duration	tinyint	NO	DF 1, CK 1–4
PitchAmount	decimal(12,2)	NO	CK ≥ 0
DiscountAmount	decimal(12,2)	NO	DF 0, CK ≥ 0
PromoCodeId	bigint	YES	FK → money.PromoCodes (NO ACTION)
ServiceFee	decimal(12,2)	NO	DF 0, CK ≥ 0
Total	decimal(12,2)	NO	CK ≥ 0
CommissionRate	decimal(5,2)	NO	snapshot, CK 0–100
PaymentId	bigint	YES	FK → money.Payments (NO ACTION)
PayAtVenue	bit	NO	DF 0
PolicyKey	varchar(20)	NO	cancellation policy snapshot
WaiverVersionId	bigint	YES	FK → content.LegalVersions (added later, NO ACTION)
Status	tinyint	NO	DF 0, CK 0–5 (Confirmed, Checked in, Played, No-show, Cancelled by customer, Cancelled by venue)
CheckedInAt	datetime2(0)	YES	
CancelledAt	datetime2(0)	YES	
RefundAmount	decimal(12,2)	NO	DF 0, CK ≥ 0 and ≤ Total
QrSecretHash	varbinary(64)	YES	NULL for package bookings (the package QR is used)

Indexes: `IX_Bookings_PitchId_Date`, `IX_Bookings_CustomerId_Date`, `IX_Bookings_Status_Date`, `IX_Bookings_PackageId`.

B59 booking.BookingSlots

One row per hour of a booking. **This is what makes double booking impossible.** | Column | Type | Null | Rules | |---|---|---|---| | Id | bigint IDENTITY | NO | PK | | BookingId | bigint | NO | FK → booking.Bookings (CASCADE) | | PitchId | bigint | NO | FK → pitch.Pitches (NO ACTION) | | Date | date | NO | | | Hour | tinyint | NO | CK 0–23 | | IsActive | bit | NO | DF 1 (set to 0 when the booking is cancelled) |

Filtered unique index `UQ_BookingSlots_PitchId_Date_Hour` ON (PitchId, Date, Hour) WHERE IsActive = 1.

B60 booking.SlotHolds

10-minute holds while a customer pays. | Column | Type | Null | Rules | |---|---|---|---| | Id | bigint IDENTITY | NO | PK | | PitchId | bigint | NO | FK → pitch.Pitches (CASCADE) | | Date | date | NO | | | Hour | tinyint | NO | CK 0–23; UQ (PitchId, Date, Hour) | | UserId | uniqueidentifier | NO | FK → auth.Users (CASCADE) | | ExpiresAt | datetime2(0) | NO | expired holds are deleted by a job |

B61 booking.QrScans

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
BookingId	bigint	YES	FK → booking.Bookings (NO ACTION)
PackageId	bigint	YES	FK → booking.Packages (NO ACTION)
ScannedById	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
PitchId	bigint	NO	FK → pitch.Pitches (NO ACTION)
ScannedAt	datetime2(0)	NO	DF SYSUTCDATETIME()
Result	tinyint	NO	CK 0–8 (Accepted, Bad signature, Other venue, Wrong day, Wrong hour, Cancelled, Already used, Package finished, Account blocked)
MatchNumber	tinyint	YES	e.g. 3 of 8 (accepted package scans)
Device	nvarchar(120)	YES	
IdempotencyKey	varchar(64)	NO	UQ — a repeated scan request is not counted twice

CK `BookingId IS NOT NULL OR PackageId IS NOT NULL`. Index: `IX_QrScans_PackageId_ScannedAt`.

B62 booking.WaitlistEntries — [B]

Column	Type	Null	Rules
PitchId	bigint	NO	FK → pitch.Pitches (CASCADE)
UserId	uniqueidentifier	NO	FK → auth.Users (CASCADE) — two cascade paths are allowed because they come from different parents
Date	date	NO	
Hour	tinyint	NO	CK 0–23; UQ (PitchId, Date, Hour, UserId)
Status	tinyint	NO	DF 0, CK 0–3 (Waiting, Free now, Booked, Expired)

Column	Type	Null	Rules
NotifiedAt	datetime2(0)	YES	

B63 booking.FreeSlots — [B]

Column	Type	Null	Rules
PitchId	bigint	NO	FK → pitch.Pitches (NO ACTION)
PostedById	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
Date	date	NO	
Hour	tinyint	NO	CK 0–23
DiscountPercent	decimal(5,2)	NO	DF 0, CK 0–90
OfferId	bigint	YES	FK → pitch.Offers (NO ACTION)
BookingId	bigint	YES	FK → booking.Bookings (NO ACTION) — set when booked
Status	tinyint	NO	DF 0, CK 0–3 (Live, Booked, Expired, Hidden by admin)

Filtered UQ (PitchId, Date, Hour) WHERE Status = 0.

B64 booking.SlotAlerts — [B]

Column	Type	Null	Rules
UserId	uniqueidentifier	NO	FK → auth.Users (CASCADE)
PitchId	bigint	YES	FK → pitch.Pitches (NO ACTION)
AreaId	int	YES	FK → geo.Areas (NO ACTION)
FromHour / ToHour	tinyint	YES	CK 0–24
IsActive	bit	NO	DF 1

CK `PitchId IS NOT NULL OR AreaId IS NOT NULL`.

B65 booking.Games — [B]

Column	Type	Null	Rules
Code	varchar(12)	NO	UQ
HostId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
BookingId	bigint	NO	UQ, FK → booking.Bookings (NO ACTION)
Title	nvarchar(100)	YES	
PlayersNeeded	tinyint	NO	CK 1–21
Positions	varchar(100)	YES	e.g. `GK,DEF`
Level	tinyint	NO	CK 0–3 (Any, Casual, Intermediate, Competitive)

Column	Type	Null	Rules
SharePrice	decimal(12,2)	NO	DF 0, CK ≥ 0
AutoAccept	bit	NO	DF 0
RequestsClosed	bit	NO	DF 0
Status	tinyint	NO	DF 0, CK 0–3 (Open, Full, Played, Cancelled)
HiddenByAdmin	bit	NO	DF 0

B66 booking.GameRequests — [B]

Column	Type	Null	Rules
GameId	bigint	NO	FK → booking.Games (CASCADE)
PlayerId	uniqueidentifier	NO	FK → auth.Users (NO ACTION); UQ (GameId, PlayerId)
Position	varchar(10)	YES	
Message	nvarchar(300)	YES	
PaymentId	bigint	YES	FK → money.Payments (NO ACTION)
Status	tinyint	NO	DF 0, CK 0–5 (Waiting, Accepted, Declined, Withdrawn, Removed, Waitlisted)

B67 booking.PlayerRatings — [B]

Column	Type	Null	Rules
GameId	bigint	NO	FK → booking.Games (CASCADE)
RaterId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
PlayerId	uniqueidentifier	NO	FK → auth.Users (NO ACTION); UQ (GameId, PlayerId)
Stars	tinyint	YES	CK 1–5
ShowedUp	bit	NO	

B68 booking.Tournaments — [B]

Column	Type	Null	Rules
PartnerId	bigint	NO	FK → pitch.Partners (NO ACTION)
PitchId	bigint	NO	FK → pitch.Pitches (NO ACTION)
Name	nvarchar(120)	NO	
Type	tinyint	NO	CK 0–1 (Knockout cup, League)
TeamCount	tinyint	NO	CK `(Type = 0 AND TeamCount IN (4,8,16)) OR (Type = 1 AND TeamCount BETWEEN 3 AND 10)`
EntryFee	decimal(12,2)	NO	DF 0, CK ≥ 0
PrizeText	nvarchar(200)	YES	

Column	Type	Null	Rules
StartsOn / EndsOn	date	NO	CK EndsOn ≥ StartsOn
MatchHours	varchar(50)	NO	e.g. `20,21`
RulesText	nvarchar(1000)	YES	
Status	tinyint	NO	DF 0, CK 0–3 (Registration, Running, Finished, Cancelled)
ChampionTeamId	bigint	YES	FK → booking.TournamentTeams (added after B69, NO ACTION)

B69 booking.TournamentTeams — [B]

Column	Type	Null	Rules
TournamentId	bigint	NO	FK → booking.Tournaments (CASCADE)
Name	nvarchar(80)	NO	UQ (TournamentId, Name)
CaptainId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
Phone	varchar(20)	NO	
PaymentId	bigint	YES	FK → money.Payments (NO ACTION)
WaiverVersionId	bigint	YES	FK → content.LegalVersions (NO ACTION)
Status	tinyint	NO	DF 0, CK 0–2 (Registered, Withdrawn, Refunded)

B70 booking.TournamentMatches — [B]

Column	Type	Null	Rules
TournamentId	bigint	NO	FK → booking.Tournaments (CASCADE)
Round	tinyint	NO	CK ≥ 1
HomeTeamId / AwayTeamId	bigint	YES	FK → booking.TournamentTeams (NO ACTION); CK Home ≠ Away
BookingId	bigint	YES	FK → booking.Bookings (NO ACTION) — the reserved hour
HomeGoals / AwayGoals	tinyint	YES	
HomePens / AwayPens	tinyint	YES	
WinnerTeamId	bigint	YES	FK → booking.TournamentTeams (NO ACTION)

B71 pitch.Reviews — [B]

Column	Type	Null	Rules
PitchId	bigint	NO	FK → pitch.Pitches (NO ACTION)
BookingId	bigint	NO	UQ, FK → booking.Bookings (NO ACTION) — one review per played booking
UserId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
Stars	tinyint	NO	CK 1–5

Column	Type	Null	Rules
Text	nvarchar(1000)	YES	
Reply	nvarchar(1000)	YES	
RepliedAt	datetime2(0)	YES	
IsHidden	bit	NO	DF 0

Index: `IX_Reviews_PitchId_CreatedAt`.

B72 pitch.ReviewReports — [B]

Column	Type	Null	Rules
ReviewId	bigint	NO	FK → pitch.Reviews (CASCADE)
ReporterId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
Reason	nvarchar(300)	NO	
Status	tinyint	NO	DF 0, CK 0–2 (Open, Hidden, Kept)

B73 shop.Orders — [B] [V]

Column	Type	Null	Rules
Code	varchar(12)	NO	UQ (e.g. `ORD-720730`)
CustomerId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
AddressJson	nvarchar(max)	NO	address snapshot, CK ISJSON
Phone	varchar(20)	NO	
Subtotal	decimal(12,2)	NO	CK ≥ 0
DiscountAmount	decimal(12,2)	NO	DF 0, CK ≥ 0
DeliveryFee	decimal(12,2)	NO	DF 0, CK ≥ 0
Total	decimal(12,2)	NO	CK ≥ 0
VatAmount	decimal(12,2)	NO	DF 0
PromoCodeId	bigint	YES	FK → money.PromoCodes (NO ACTION)
PaymentMethod	tinyint	NO	CK 0–2 (Online, Cash on delivery, Wallet)
PaymentId	bigint	YES	FK → money.Payments (NO ACTION)
Status	tinyint	NO	DF 0, CK 0–5 (Placed, Paid, In progress, Completed, Cancelled, Refunded)
DeliverySpeed	tinyint	NO	DF 0, CK 0–1 (Standard, Express)

Index: `IX_Orders_CustomerId_CreatedAt`, `IX_Orders_Status_CreatedAt`.

B74 shop.OrderItems — [B]

Column	Type	Null	Rules
OrderId	bigint	NO	FK → shop.Orders (CASCADE)
Kind	tinyint	NO	CK 0–1 (Store product, Printed design)
StoreId	bigint	NO	FK → shop.Stores (NO ACTION)
ProductVariantId	bigint	YES	FK → shop.ProductVariants (NO ACTION)
DesignId	bigint	YES	FK → pod.Designs (NO ACTION)
Title	nvarchar(150)	NO	snapshot
Size / Color	nvarchar(40)	NO	
Qty	int	NO	CK 1–20
UnitPrice	decimal(12,2)	NO	CK ≥ 0
BaseAmount	decimal(12,2)	YES	printed items: Super App base per unit (snapshot)
CommissionAmount	decimal(12,2)	NO	DF 0
SellerEarning	decimal(12,2)	NO	DF 0
Status	tinyint	NO	DF 0, CK 0–7 (Received, Printing, Packed, Shipped, Delivered, Refused, Cancelled, Returned)
DeliveredAt	datetime2(0)	YES	

CK `(Kind = 0 AND ProductVariantId IS NOT NULL AND DesignId IS NULL) OR (Kind = 1 AND DesignId IS NOT NULL AND ProductVariantId IS NULL)`. Indexes: `IX_OrderItems_OrderId`, `IX_OrderItems_StoreId_Status`, `IX_OrderItems_DesignId`.

B75 shop.Fulfilments — [B]

One per store per order (each store ships its part; printed items are shipped by the factory). | Column | Type | Null | Rules | |---|---|---|---| | OrderId | bigint | NO | FK → shop.Orders (CASCADE) | | StoreId | bigint | NO | FK → shop.Stores (NO ACTION); UQ (OrderId, StoreId) | | Status | tinyint | NO | DF 0, CK 0–4 (New, Packed, Shipped, Delivered, Cancelled) |

B76 shop.CartItems — [B]

Column	Type	Null	Rules
UserId	uniqueidentifier	NO	FK → auth.Users (CASCADE)
ProductVariantId	bigint	YES	FK → shop.ProductVariants (NO ACTION)
DesignId	bigint	YES	FK → pod.Designs (NO ACTION)
Size / Color	nvarchar(40)	YES	printed items
Qty	int	NO	CK 1–20

CK exactly one of ProductVariantId / DesignId. UQ filtered per (UserId, ProductVariantId) and per (UserId, DesignId, Size, Color).

B77 factory.Jobs — [B] [V]

Column	Type	Null	Rules
Code	varchar(16)	NO	UQ (printed on the ticket QR)
OrderItemId	bigint	NO	UQ, FK → shop.OrderItems (NO ACTION)
FactoryId	bigint	NO	FK → factory.Factories (NO ACTION)
Stage	tinyint	NO	DF 0, CK 0–6 (New, Printing, Quality check, Packing, Ready, Shipped, Cancelled)
DueAt	datetime2(0)	NO	
AssignedStaffId	bigint	YES	FK → factory.FactoryStaff (NO ACTION)
ReprintCount	tinyint	NO	DF 0
IsSample	bit	NO	DF 0
DeductionAmount	decimal(12,2)	NO	DF 0, CK ≥ 0

Indexes: `IX_Jobs_FactoryId_Stage_DueAt`.

B78 factory.JobEvents

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
JobId	bigint	NO	FK → factory.Jobs (CASCADE)
FromStage / ToStage	tinyint	NO	CK 0–6
StaffId	bigint	YES	FK → factory.FactoryStaff (NO ACTION)
Note	nvarchar(300)	YES	e.g. QC fail reason
At	datetime2(0)	NO	DF SYSUTCDATETIME()

B79 factory.JobProblems — [B]

Column	Type	Null	Rules
JobId	bigint	NO	FK → factory.Jobs (CASCADE)
ReportedByStaffId	bigint	YES	FK → factory.FactoryStaff (NO ACTION)
Kind	tinyint	NO	CK 0–4 (Out of blanks, Machine down, Bad artwork, Damaged, Other)
Detail	nvarchar(400)	NO	
Status	tinyint	NO	DF 0, CK 0–1 (Open, Resolved)
ResolvedById	uniqueidentifier	YES	FK → auth.Users (NO ACTION)
Resolution	nvarchar(400)	YES	

B80 pod.Samples — [B]

Column	Type	Null	Rules
SellerId	bigint	NO	FK → pod.Sellers (NO ACTION)
DesignId	bigint	NO	FK → pod.Designs (NO ACTION)
Color / Size	nvarchar(40)	NO	
Address	nvarchar(400)	NO	
Price	decimal(12,2)	NO	CK ≥ 0
IsFree	bit	NO	DF 0
OrderId	bigint	YES	FK → shop.Orders (NO ACTION) — created when approved
Status	tinyint	NO	DF 0, CK 0–4 (Requested, Approved, Rejected, Shipped, Delivered)

B81 ship.Shipments — [B] [V]

Column	Type	Null	Rules
OrderId	bigint	NO	FK → shop.Orders (NO ACTION)
FulfilmentId	bigint	YES	FK → shop.Fulfilments (NO ACTION); NULL = printed items from the factory
FactoryId	bigint	YES	FK → factory.Factories (NO ACTION)
Provider	varchar(40)	NO	integration key
Mode	tinyint	NO	CK 0–2 (Simulated, Test, Live)
TrackingNumber	varchar(60)	NO	UQ (Provider, TrackingNumber)
CodAmount	decimal(12,2)	NO	DF 0, CK ≥ 0
LabelFileId	bigint	YES	FK → ops.Files (NO ACTION)
Status	tinyint	NO	DF 0, CK 0–6 (Created, Awaiting pickup, Picked up, Out for delivery, Delivered, Refused, Cancelled)
PickupAt / DeliveredAt	datetime2(0)	YES	
CodRemittedAt	datetime2(0)	YES	when the courier paid us the cash
IsReturn	bit	NO	DF 0 — reverse pickup

B82 ship.ShipmentEvents

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
ShipmentId	bigint	NO	FK → ship.Shipments (CASCADE)
Status	tinyint	NO	CK 0–6
Description	nvarchar(300)	YES	

Column	Type	Null	Rules
RawJson	nvarchar(max)	YES	the courier's webhook, CK ISJSON
ProviderEventId	varchar(80)	YES	UQ filtered — replayed webhooks are ignored
At	datetime2(0)	NO	

B83 ship.Returns — [B]

Column	Type	Null	Rules
Code	varchar(14)	NO	UQ (e.g. `RT-AS5CP26`)
OrderId	bigint	NO	FK → shop.Orders (NO ACTION)
CustomerId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
Reason	tinyint	NO	CK 0–4 (Wrong size, Faulty or damaged, Wrong item or print, Not as described, Changed my mind)
Detail	nvarchar(600)	YES	
Status	tinyint	NO	DF 0, CK 0–6 (Requested, Approved, Received, Refunded, Exchanged, Reprinting, Rejected)
DecidedById	uniqueidentifier	YES	FK → auth.Users (NO ACTION)
RejectReason	nvarchar(400)	YES	
RefundAmount	decimal(12,2)	NO	DF 0, CK ≥ 0
CustomerPaysDelivery	bit	NO	DF 0

B84 ship.ReturnItems

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
ReturnId	bigint	NO	FK → ship.Returns (CASCADE)
OrderItemId	bigint	NO	FK → shop.OrderItems (NO ACTION)
Qty	int	NO	CK ≥ 1
Action	tinyint	NO	CK 0–2 (Refund, Exchange, Reprint)
NewSize / NewColor	nvarchar(40)	YES	exchanges

B85 money.LedgerEntries

Append-only. Every money movement is a row; balances are sums. | Column | Type | Null | Rules | |---|---|---|---| | Id | bigint IDENTITY | NO | PK | | WalletId | bigint | NO | FK → money.Wallets (NO ACTION) | | Amount | decimal(14,2) | NO | CK ≠ 0 (positive = in, negative = out) | | Bucket | tinyint | NO | CK 0–2 (Pending, Available, Settled) | | Kind | tinyint | NO | CK 0–14 (Booking earning, Store earning, POD profit, Commission, Service fee, Refund, Payout, Adjustment, Release from hold, Factory earning, Deduction, Voucher, Points redemption, **Wallet top-up**, **Wallet payment**) | | RefType | varchar(30) | NO | e.g. `booking`, `orderItem`, `payout` | | RefId |

bigint | NO | | | AvailableAt | datetime2(0) | YES | when a pending amount becomes available (holds) | | GroupId | uniqueidentifier | NO | the entries of one transaction share it (they always sum to zero across wallets) | | CreatedAt | datetime2(0) | NO | DF SYSUTCDATETIME() |

Indexes: `IX_Ledger_WalletId_CreatedAt`, `IX_Ledger_Bucket_AvailableAt` (release job), `IX_Ledger_RefType_RefId`.

B86 money.Refunds — [B]

Column	Type	Null	Rules
PaymentId	bigint	NO	FK → money.Payments (NO ACTION)
Amount	decimal(12,2)	NO	CK > 0
Destination	tinyint	NO	CK 0–1 (Original method, Super App wallet)
PaidBy	tinyint	NO	CK 0–3 (Super App, Partner, Seller, Factory)
Reason	nvarchar(400)	NO	
ProviderRef	varchar(100)	YES	
Status	tinyint	NO	DF 0, CK 0–3 (Requested, Waiting for approval, Done, Failed)
RequestedById	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
ApprovalId	bigint	YES	FK → money.Approvals (added after B88, NO ACTION)

B87 money.PayoutRequests — [B] [V]

Column	Type	Null	Rules
Code	varchar(14)	NO	UQ
WalletId	bigint	NO	FK → money.Wallets (NO ACTION)
Amount	decimal(12,2)	NO	CK > 0 (minimum 500 checked from settings)
Method	tinyint	NO	CK 0–1 (Bank, InstaPay)
Destination	nvarchar(200)	NO	encrypted snapshot
Status	tinyint	NO	DF 0, CK 0–5 (Requested, Prepared, Waiting for approval, Sent, Rejected, Failed)
PreparedById	uniqueidentifier	YES	FK → auth.Users (NO ACTION)
SentById	uniqueidentifier	YES	FK → auth.Users (NO ACTION); CK `SentById <> PreparedById`
TransferRef	varchar(100)	YES	CK required when Status = 3
RejectReason	nvarchar(400)	YES	CK required when Status = 4
ApprovalId	bigint	YES	FK → money.Approvals (added after B88, NO ACTION)
SentAt	datetime2(0)	YES	

B88 money.Approvals — [B]

Everything over a limit waits here (CEO inbox). | Column | Type | Null | Rules | |---|---|---|---| | Kind | tinyint | NO | CK 0–5 (Payout, Refund, Credit, Settlement, Package discount over policy, Other) | | RefType | varchar(30) | NO | | | RefId | bigint | NO | | | Amount | decimal(14,2) | NO | DF 0, CK ≥ 0 | | Summary | nvarchar(300) | NO | | | RequestedById | uniqueidentifier | NO | FK → auth.Users (NO ACTION) | | RequiredApprovals | tinyint | NO | DF 1, CK 1–2 (two above 50,000 EGP) | | Status | tinyint | NO | DF 0, CK 0–3 (Waiting, Approved, Rejected, Cancelled) | | DecidedAt | datetime2(0) | YES | |

Index: `IX_Approvals_Status_CreatedAt`.

B89 money.ApprovalDecisions — [B]

Column	Type	Null	Rules
ApprovalId	bigint	NO	FK → money.Approvals (CASCADE)
DeciderId	uniqueidentifier	NO	FK → auth.Users (NO ACTION); UQ (ApprovalId, DeciderId) — one person cannot give both approvals
Decision	tinyint	NO	CK 0–1 (Approve, Reject)
Note	nvarchar(400)	YES	

B90 money.Adjustments — [B]

Column	Type	Null	Rules
WalletId	bigint	NO	FK → money.Wallets (NO ACTION)
Amount	decimal(12,2)	NO	CK ≠ 0
Reason	nvarchar(400)	NO	
ById	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
ApprovalId	bigint	YES	FK → money.Approvals (NO ACTION)

B91 money.Receipts — [B]

Column	Type	Null	Rules
Number	varchar(20)	NO	UQ (sequential per year, e.g. `R-2026-000123`)
UserId	uniqueidentifier	YES	FK → auth.Users (NO ACTION)
Kind	tinyint	NO	CK 0–6 (Payment, Refund, Payout advice, Top-up, Voucher, Statement, Tax invoice)
Amount	decimal(12,2)	NO	CK ≥ 0
VatAmount	decimal(12,2)	NO	DF 0
RefType / RefId	varchar(30) / bigint	NO	
FileId	bigint	YES	FK → ops.Files (NO ACTION)

Column	Type	Null	Rules
EtaUuid	varchar(64)	YES	ETA e-receipt id (later)

B92 money.PromoCodeUses — [B]

Column	Type	Null	Rules
PromoCodeId	bigint	NO	FK → money.PromoCodes (NO ACTION)
UserId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
OrderId	bigint	YES	FK → shop.Orders (NO ACTION)
BookingId	bigint	YES	FK → booking.Bookings (NO ACTION)
DiscountAmount	decimal(12,2)	NO	CK > 0
RevenueAmount	decimal(12,2)	NO	DF 0

CK `OrderId IS NOT NULL OR BookingId IS NOT NULL`. Index: `IX_PromoCodeUses_PromoCodeId_UserId`.

B93 money.PointEntries

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
UserId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
Points	int	NO	CK ≠ 0
Kind	tinyint	NO	CK 0–5 (Booking, Order, Tournament, Review, Voucher redemption, Admin adjustment)
RefType / RefId	varchar(30) / bigint	YES	
Reason	nvarchar(200)	YES	required for admin adjustments
CreatedAt	datetime2(0)	NO	DF SYSUTCDATETIME()

Index: `IX_PointEntries_UserId_CreatedAt`.

B94 money.Vouchers — [B]

Column	Type	Null	Rules
Code	varchar(20)	NO	UQ
UserId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
Amount	decimal(12,2)	NO	CK ≥ 100
Source	tinyint	NO	CK 0–2 (Points, Fantasy prize, Admin)
PointsUsed	int	NO	DF 0, CK ≥ 0
Status	tinyint	NO	DF 0, CK 0–2 (Active, Used, Expired)

Column	Type	Null	Rules
UsedOnType / UsedOnId	varchar(30) / bigint	YES	
ExpiresAt	datetime2(0)	YES	

7. Part B — Child tables: fantasy, communities, support, content, notifications

B95 fantasy.Seasons — [B]

Column	Type	Null	Rules
CompetitionId	int	NO	FK → fantasy.Competitions (NO ACTION)
Year	smallint	NO	UQ (CompetitionId, Year), CK 2020–2100
IsCurrent	bit	NO	DF 0 — filtered UQ (CompetitionId) WHERE IsCurrent = 1
LastSyncedAt	datetime2(0)	YES	

B96 fantasy.Players — [B]

Column	Type	Null	Rules
ExternalId	int	NO	UQ — API-FOOTBALL player id
ClubId	bigint	NO	FK → fantasy.Clubs (NO ACTION)
Name	nvarchar(100)	NO	real player name
NameAr	nvarchar(100)	YES	
Position	tinyint	NO	CK 0–3 (GK, DEF, MID, FWD)
Price	decimal(4,1)	NO	CK 3.5–15.0
PhotoUrl	nvarchar(300)	YES	
Status	tinyint	NO	DF 0, CK 0–3 (Available, Doubtful, Injured, Left the club)
SelectedByPercent	decimal(5,2)	NO	DF 0

Index: `IX_Players_ClubId_Position`.

B97 fantasy.Gameweeks — [B]

Column	Type	Null	Rules
SeasonId	bigint	NO	FK → fantasy.Seasons (CASCADE)
Number	tinyint	NO	UQ (SeasonId, Number), CK 1–50
Deadline	datetime2(0)	NO	
Status	tinyint	NO	DF 0, CK 0–3 (Upcoming, Live, Scoring, Finished)
ScoredAt	datetime2(0)	YES	

B98 fantasy.Fixtures — [B]

Column	Type	Null	Rules
ExternalId	int	NO	UQ
GameweekId	bigint	NO	FK → fantasy.Gameweeks (CASCADE)
HomeClubId / AwayClubId	bigint	NO	FK → fantasy.Clubs (NO ACTION); CK Home ≠ Away
KickoffAt	datetime2(0)	NO	
Status	tinyint	NO	DF 0, CK 0–4 (Scheduled, Live, Finished, Postponed, Cancelled)
HomeGoals / AwayGoals	tinyint	YES	
Minute	tinyint	YES	live minute

Index: `IX_Fixtures_GameweekId`, `IX_Fixtures_KickoffAt`.

B99 fantasy.PlayerFixtureStats — [B]

Column	Type	Null	Rules
PlayerId	bigint	NO	FK → fantasy.Players (NO ACTION)
FixtureId	bigint	NO	FK → fantasy.Fixtures (CASCADE); UQ (PlayerId, FixtureId)
Minutes	tinyint	NO	DF 0, CK 0–130
Goals / Assists / Saves / OwnGoals / PenaltiesSaved / PenaltiesMissed	tinyint	NO	DF 0
YellowCards	tinyint	NO	DF 0, CK 0–2
RedCard	bit	NO	DF 0
CleanSheet	bit	NO	DF 0
GoalsConceded	tinyint	NO	DF 0
Bonus	tinyint	NO	DF 0, CK 0–3
Points	smallint	NO	DF 0 — calculated by the scoring job

B100 fantasy.FantasyTeams — [B] [V]

Column	Type	Null	Rules
UserId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
SeasonId	bigint	NO	FK → fantasy.Seasons (NO ACTION); UQ (UserId, SeasonId)
Name	nvarchar(40)	NO	
Bank	decimal(5,1)	NO	DF 0, CK ≥ 0
FreeTransfers	tinyint	NO	DF 1, CK 0–5

Column	Type	Null	Rules
TotalPoints	int	NO	DF 0
OverallRank	int	YES	
ChipsUsed	varchar(60)	YES	e.g. `wildcard1,benchboost`

B101 fantasy.Picks

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
TeamId	bigint	NO	FK → fantasy.FantasyTeams (CASCADE)
GameweekId	bigint	NO	FK → fantasy.Gameweeks (NO ACTION)
PlayerId	bigint	NO	FK → fantasy.Players (NO ACTION); UQ (TeamId, GameweekId, PlayerId)
Slot	tinyint	NO	CK 1–15 (12–15 = bench); UQ (TeamId, GameweekId, Slot)
IsCaptain / IsViceCaptain	bit	NO	DF 0
Multiplier	tinyint	NO	DF 1, CK 0–3
Points	smallint	NO	DF 0

B102 fantasy.Transfers — [B]

Column	Type	Null	Rules
TeamId	bigint	NO	FK → fantasy.FantasyTeams (CASCADE)
GameweekId	bigint	NO	FK → fantasy.Gameweeks (NO ACTION)
PlayerOutId / PlayerInId	bigint	NO	FK → fantasy.Players (NO ACTION); CK Out ≠ In
PriceOut / PriceIn	decimal(4,1)	NO	
PointsHit	tinyint	NO	DF 0 (4 per extra transfer)

B103 fantasy.TeamGameweekPoints — [B]

Column	Type	Null	Rules
TeamId	bigint	NO	FK → fantasy.FantasyTeams (CASCADE)
GameweekId	bigint	NO	FK → fantasy.Gameweeks (NO ACTION); UQ (TeamId, GameweekId)
Points	smallint	NO	DF 0
TransfersCost	tinyint	NO	DF 0
Rank	int	YES	

Column	Type	Null	Rules
ChipPlayed	varchar(20)	YES	

Index: `IX_TeamGameweekPoints_GameweekId_Points` (rankings).

B104 fantasy.FantasyGroups — [B] [S]

Column	Type	Null	Rules
SeasonId	bigint	NO	FK → fantasy.Seasons (NO ACTION)
Name	nvarchar(60)	NO	
InviteCode	varchar(10)	NO	UQ
Type	tinyint	NO	CK 0–2 (Official, Private, Paid)
IsHidden	bit	NO	DF 0 — **paid groups stay hidden until a payment gateway is live**
MonthlyPrice	decimal(12,2)	YES	CK > 0; CK required when Type = 2
OwnerUserId	uniqueidentifier	YES	FK → auth.Users (NO ACTION)
SponsorId	bigint	YES	FK → fantasy.Sponsors (NO ACTION)
PrizeText	nvarchar(200)	YES	sponsor rewards / vouchers only — never cash
PrizeValue	decimal(12,2)	NO	DF 0

B105 fantasy.FantasyPasses — [B]

Column	Type	Null	Rules
UserId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
CompetitionId	int	YES	FK → fantasy.Competitions (NO ACTION); NULL = both leagues
PaymentId	bigint	YES	FK → money.Payments (NO ACTION)
Price	decimal(12,2)	NO	CK ≥ 0 (free for Elite sellers)
StartsAt / EndsAt	datetime2(0)	NO	CK EndsAt > StartsAt

Index: `IX_FantasyPasses_UserId_EndsAt`.

B106 community.Communities — [B] [A]

Column	Type	Null	Rules
Slug	varchar(40)	NO	UQ
Name	nvarchar(80)	NO	
About	nvarchar(500)	YES	
Audience	tinyint	NO	CK 0–2 (Everyone, Sellers, Pro & Elite sellers)
RulesText	nvarchar(1000)	YES	

Column	Type	Null	Rules
SlowModeSeconds	int	NO	DF 0, CK 0–3600
Color / Icon	varchar(30)	YES	
IsArchived	bit	NO	DF 0

B107 community.Posts — [B] [S]

Column	Type	Null	Rules
CommunityId	bigint	NO	FK → community.Communities (NO ACTION)
AuthorId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
AsSuperAppAdmin	bit	NO	DF 0
Text	nvarchar(2000)	NO	CK LEN ≥ 1
ImageFileId	bigint	YES	FK → ops.Files (NO ACTION)
HasPoll	bit	NO	DF 0
IsPinned / IsAnnouncement / CommentsClosed / IsHidden	bit	NO	DF 0
CommentsCount / ReactionsCount	int	NO	DF 0 (cached)

Index: `IX_Posts_CommunityId_CreatedAt`, `IX_Posts_AuthorId`.

B108 community.PollOptions

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
PostId	bigint	NO	FK → community.Posts (CASCADE)
Text	nvarchar(80)	NO	
SortOrder	tinyint	NO	DF 0, CK 0–5
VotesCount	int	NO	DF 0

B109 community.Comments — [B] [S]

Column	Type	Null	Rules
PostId	bigint	NO	FK → community.Posts (CASCADE)
ParentCommentId	bigint	YES	FK → community.Comments (NO ACTION, self) — replies
AuthorId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
AsSuperAppAdmin	bit	NO	DF 0
Text	nvarchar(1000)	NO	

Column	Type	Null	Rules
IsHidden	bit	NO	DF 0

B110 community.Reactions — [B]

Column	Type	Null	Rules
UserId	uniqueidentifier	NO	FK → auth.Users (CASCADE)
PostId	bigint	YES	FK → community.Posts (NO ACTION)
CommentId	bigint	YES	FK → community.Comments (NO ACTION)
Type	tinyint	NO	CK 0–5 (Like, Fire, Goal, Laugh, Clap, Sad)

CK exactly one target. Filtered UQ (UserId, PostId) and (UserId, CommentId).

B111 community.ContentReports — [B]

Column	Type	Null	Rules
ReporterId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
PostId	bigint	YES	FK → community.Posts (NO ACTION)
CommentId	bigint	YES	FK → community.Comments (NO ACTION)
Reason	tinyint	NO	CK 0–4 (Spam, Abuse, Scam, Off-topic, Other)
Detail	nvarchar(400)	YES	
Status	tinyint	NO	DF 0, CK 0–3 (Open, Removed, Kept, Removed and author muted)
HandledById	uniqueidentifier	YES	FK → auth.Users (NO ACTION)

CK exactly one target.

B112 support.Tickets — [B] [V]

Column	Type	Null	Rules
Code	varchar(10)	NO	UQ (e.g. `TK-2054`)
OpenedById	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
OpenedFrom	tinyint	NO	CK 0–3 (Site, Partner dashboard, Seller dashboard, Factory)
Topic	tinyint	NO	CK 0–7 (Booking, Order, Payment, Account, Fantasy, Seller, Partner, Other)
Subject	nvarchar(150)	NO	
BookingId	bigint	YES	FK → booking.Bookings (SET NULL)
OrderId	bigint	YES	FK → shop.Orders (SET NULL)
AgainstUserId	uniqueidentifier	YES	FK → auth.Users (NO ACTION) — dispute counterparty

Column	Type	Null	Rules
IsDispute	bit	NO	DF 0
DisputeAmount	decimal(12,2)	YES	CK > 0
Priority	tinyint	NO	DF 1, CK 0–3 (Low, Normal, High, Urgent)
Status	tinyint	NO	DF 0, CK 0–4 (Open, Waiting for customer, Waiting for team, Resolved, Closed)
AssignedToUserId	uniqueidentifier	YES	FK → auth.Users (NO ACTION)
AssignedToAgentId	bigint	YES	FK → ai.Agents (NO ACTION)
FirstResponseAt / ResolvedAt	datetime2(0)	YES	

Indexes: `IX_Tickets_Status_Priority_CreatedAt`, `IX_Tickets_OpenedById`.

B113 support.TicketMessages — [B]

Column	Type	Null	Rules
TicketId	bigint	NO	FK → support.Tickets (CASCADE)
SenderId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
Body	nvarchar(4000)	NO	CK LEN ≥ 1
IsInternalNote	bit	NO	DF 0 — visible to the team only
ReadAt	datetime2(0)	YES	

Index: `IX_TicketMessages_TicketId_CreatedAt`.

B114 support.TicketAttachments

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
MessageId	bigint	NO	FK → support.TicketMessages (CASCADE)
FileId	bigint	NO	FK → ops.Files (NO ACTION)

B115 content.HelpArticles — [B]

Column	Type	Null	Rules
TopicId	int	NO	FK → content.HelpTopics (NO ACTION)
QuestionEn / QuestionAr	nvarchar(200)	NO	
AnswerEn / AnswerAr	nvarchar(4000)	NO	
SortOrder	int	NO	DF 0
HelpfulCount / NotHelpfulCount	int	NO	DF 0

Column	Type	Null	Rules
IsPublished	bit	NO	DF 1

B116 content.HelpVotes — [B]

Column	Type	Null	Rules
ArticleId	bigint	NO	FK → content.HelpArticles (CASCADE)
UserId	uniqueidentifier	YES	FK → auth.Users (SET NULL)
VisitorId	varchar(40)	YES	anonymous visitors
Helpful	bit	NO	

B117 content.LegalVersions — [B]

Column	Type	Null	Rules
DocumentId	int	NO	FK → content.LegalDocuments (NO ACTION)
Version	smallint	NO	UQ (DocumentId, Version), CK ≥ 1
BodyEn / BodyAr	nvarchar(max)	NO	
SummaryEn / SummaryAr	nvarchar(400)	NO	
PublishedAt	datetime2(0)	YES	
LawyerReviewedAt	datetime2(0)	YES	the "Draft" note disappears when set
LawyerNote	nvarchar(400)	YES	
ReviewedById	uniqueidentifier	YES	FK → auth.Users (NO ACTION)

After this table exists, `auth.Applications.AgreementVersionId`, `booking.Bookings.WaiverVersionId` and `booking.TournamentTeams.WaiverVersionId` get their FKs.

B118 content.LegalAcceptances

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
VersionId	bigint	NO	FK → content.LegalVersions (NO ACTION)
UserId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
Context	varchar(30)	NO	`signup`, `booking`, `game`, `tournament`, `seller`, `partner`
ContextId	bigint	YES	
Ip	varchar(45)	YES	
AcceptedAt	datetime2(0)	NO	DF SYSUTCDATETIME()

Index: `IX_LegalAcceptances_UserId_VersionId`.

B119 content.CookieConsents

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
VisitorId	varchar(40)	NO	
UserId	uniqueidentifier	YES	FK → auth.Users (SET NULL)
Analytics	bit	NO	
PolicyVersion	smallint	NO	
At	datetime2(0)	NO	DF SYSUTCDATETIME()

B120 content.NewsPosts — [B] [S]

Column	Type	Null	Rules
AuthorId	uniqueidentifier	NO	FK → auth.Users (NO ACTION)
Category	varchar(20)	NO	
Title	nvarchar(110)	NO	CK LEN ≥ 10
Excerpt	nvarchar(260)	NO	CK LEN ≥ 20
Body	nvarchar(max)	NO	
ImageFileId	bigint	YES	FK → ops.Files (NO ACTION)
IsBreaking / IsHidden	bit	NO	DF 0
PublishedAt	datetime2(0)	NO	DF SYSUTCDATETIME()

B121 notify.Notifications

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
UserId	uniqueidentifier	NO	FK → auth.Users (CASCADE)
Category	tinyint	NO	CK 0–10 (Bookings, Orders, Designs, Money, Level, Store, Account, Community, Support, Fantasy, System)
Title	nvarchar(150)	NO	
Body	nvarchar(500)	YES	
Link	nvarchar(300)	YES	CK `Link LIKE '#/%'`
Tone	tinyint	NO	DF 0, CK 0–3 (Info, Success, Warning, Danger)
ReadAt	datetime2(0)	YES	
CreatedAt	datetime2(0)	NO	DF SYSUTCDATETIME()

Index: `IX_Notifications_UserId_ReadAt_CreatedAt`. A job removes read notifications older than 180 days.

B122 notify.NotificationPreferences

Column	Type	Null	Rules
UserId	uniqueidentifier	NO	PK part, FK → auth.Users (CASCADE)
Category	tinyint	NO	PK part (UserId, Category)
InApp	bit	NO	DF 1
Email	bit	NO	DF 1
Push	bit	NO	DF 0

Account and money categories cannot be muted (checked in the service).

B123 notify.EmailOutbox

Column	Type	Null	Rules
Id	bigint IDENTITY	NO	PK
UserId	uniqueidentifier	YES	FK → auth.Users (SET NULL)
ToAddress	nvarchar(256)	NO	
Subject	nvarchar(200)	NO	
Template	varchar(60)	NO	e.g. `booking-confirmed`
ModelJson	nvarchar(max)	NO	CK ISJSON
Language	char(2)	NO	DF `'ar'`
Status	tinyint	NO	DF 0, CK 0–3 (Queued, Sent, Failed, Cancelled)
Attempts	tinyint	NO	DF 0
SentAt	datetime2(0)	YES	
Error	nvarchar(1000)	YES	
CreatedAt	datetime2(0)	NO	DF SYSUTCDATETIME()

Index: `IX_EmailOutbox_Status_CreatedAt`.

B124 notify.Broadcasts — [B] [A]

Column	Type	Null	Rules
Title	nvarchar(150)	NO	
Body	nvarchar(1000)	NO	
Link	nvarchar(300)	YES	
Segment	tinyint	NO	CK 0–5 (Everyone, Customers, Partners, Sellers, Factories, Fantasy players)
SendEmail	bit	NO	DF 0

Column	Type	Null	Rules
ScheduledAt	datetime2(0)	YES	
SentAt	datetime2(0)	YES	
RecipientsCount	int	NO	DF 0

8. Part C — Junction tables

Each junction table has a **composite primary key** of its two foreign keys (no surrogate id), plus `CreatedAt datetime2(0) NOT NULL DEFAULT SYSUTCDATETIME()`. Both foreign keys **CASCADE** unless noted (deleting either side removes the link only).

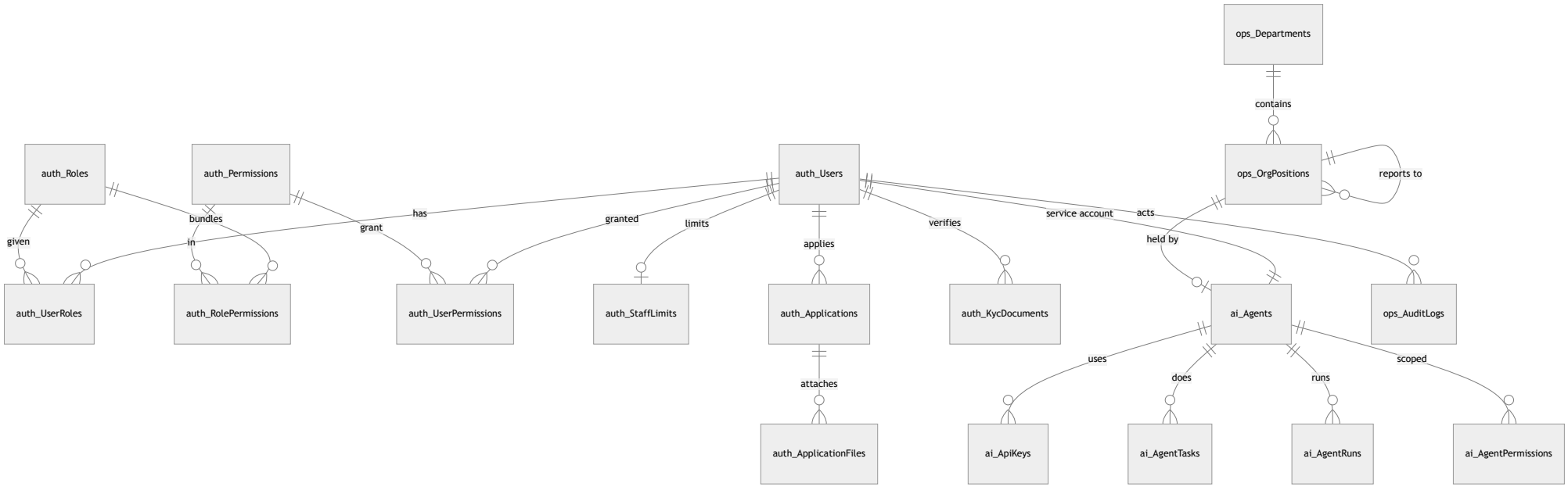
#	Table	Columns (PK)	Extra columns	Notes
C1	`auth.UserRoles`	UserId → Users, RoleId → Roles	—	Identity `AspNetUserRoles`
C2	`auth.RolePermissions`	RoleId → Roles, PermissionId → Permissions (NO ACTION)	—	edited in the CEO permissions matrix
C3	`auth.UserPermissions`	UserId → Users, PermissionId → Permissions (NO ACTION)	`GrantedById uniqueidentifier NOT NULL` (FK → Users, NO ACTION), `ExpiresAt datetime2(0) NULL`	direct grants, e.g. Approvals to one admin
C4	`ai.AgentPermissions`	AgentId → Agents, PermissionId → Permissions (NO ACTION)	—	the agent's scope
C5	`pitch.PitchFacilities`	PitchId → Pitches, FacilityId → Facilities (NO ACTION)	—	
C6	`pod.DesignColors`	DesignId → Designs, CatalogColorId → CatalogColors (NO ACTION)	—	colours the customer can choose
C7	`pod.SellerFollowers`	SellerId → Sellers, UserId → Users (NO ACTION)	—	follow a store
C8	`shop.WishlistItems`	Id bigint PK (exception), UserId → Users, ProductId → Products (NO ACTION) NULL, DesignId → Designs (NO ACTION) NULL	—	CK exactly one of ProductId / DesignId; filtered UQ per pair
C9	`money.PromoCodePitches`	PromoCodeId → PromoCodes, PitchId → Pitches (NO ACTION)	—	a booking code limited to chosen pitches

#	Table	Columns (PK)	Extra columns	Notes
C10	`fantasy.SeasonClubs`	SeasonId → Seasons, ClubId → Clubs (NO ACTION)	—	clubs playing in a season
C11	`fantasy.GroupMembers`	GroupId → FantasyGroups, TeamId → FantasyTeams (NO ACTION)	`Rank int NULL`, `PaidUntil datetime2(0) NULL` (paid groups)	
C12	`community.CommunityMembers`	CommunityId → Communities, UserId → Users (NO ACTION)	`Role tinyint NOT NULL DEFAULT 0` CK 0–2 (Member, Moderator, Owner), `MutedUntil datetime2(0) NULL`, `MuteReason nvarchar(300) NULL`	
C13	`community.PollVotes`	OptionId → PollOptions, UserId → Users (NO ACTION)	`PostId bigint NOT NULL`	**UQ (PostId, UserId)** — one vote per poll

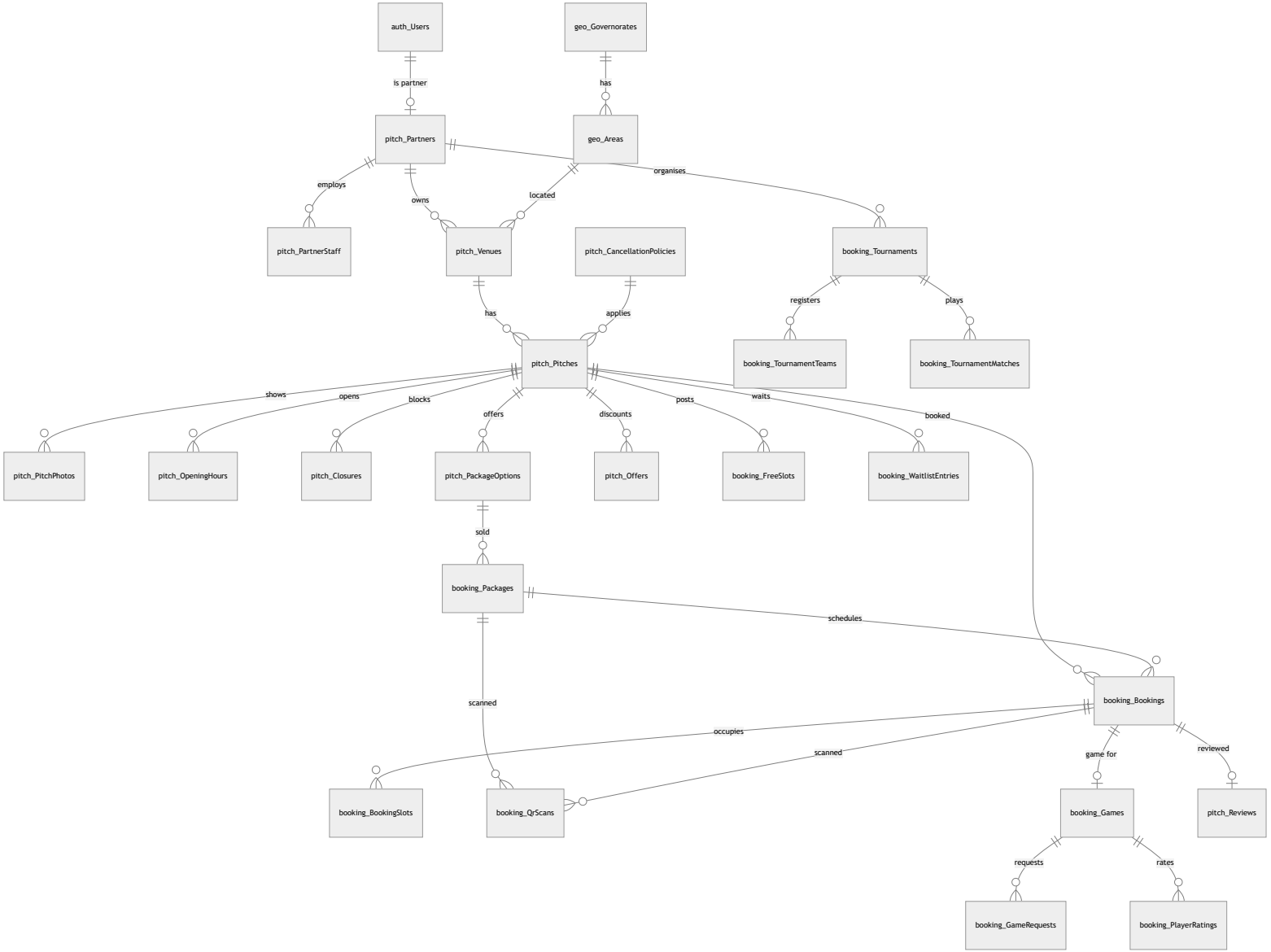
9. Relationship diagrams

The diagrams show tables and relationships (not every column). Read `||--o{` as "one to many", `||--||` as "one to one", `}o--o{` as "many to many through a junction table". Schema prefixes use `_` instead of `.`.

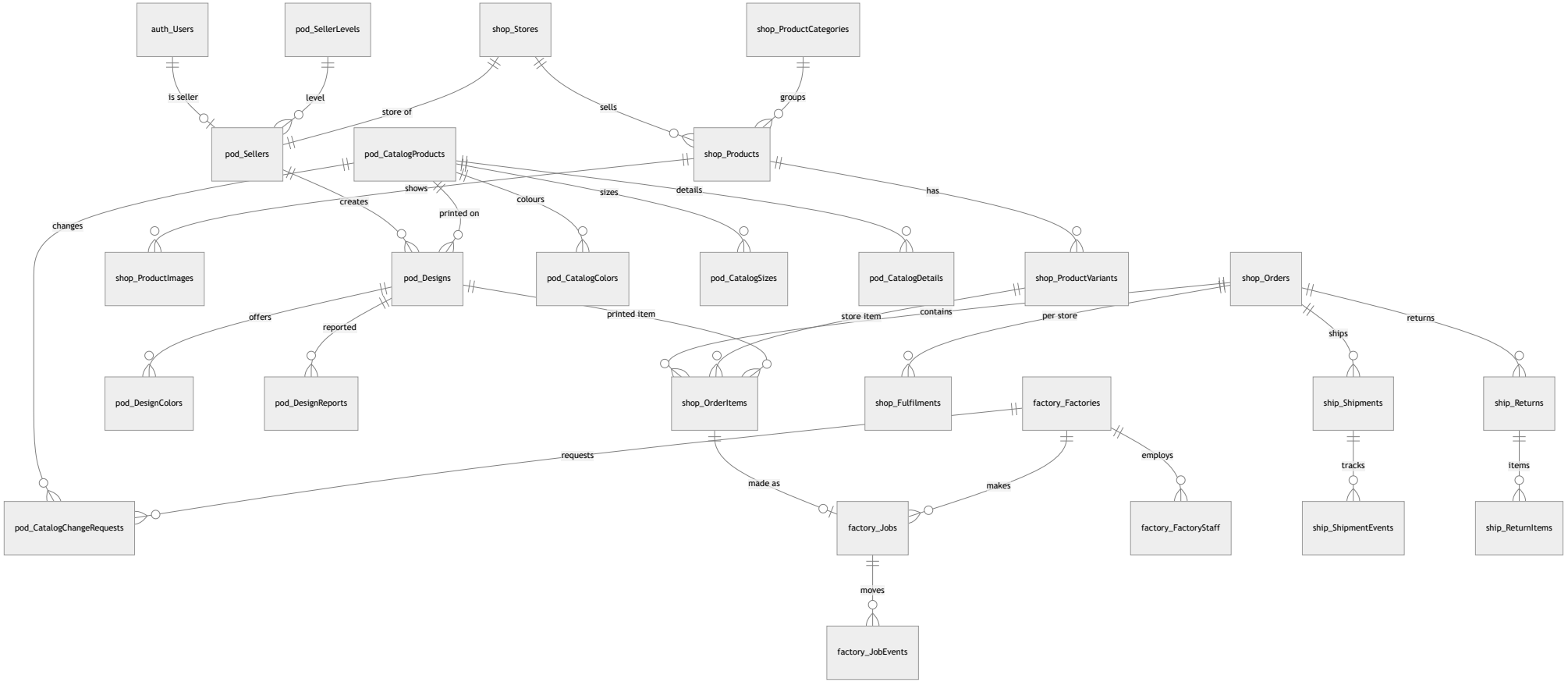
9.1 Accounts, permissions and AI



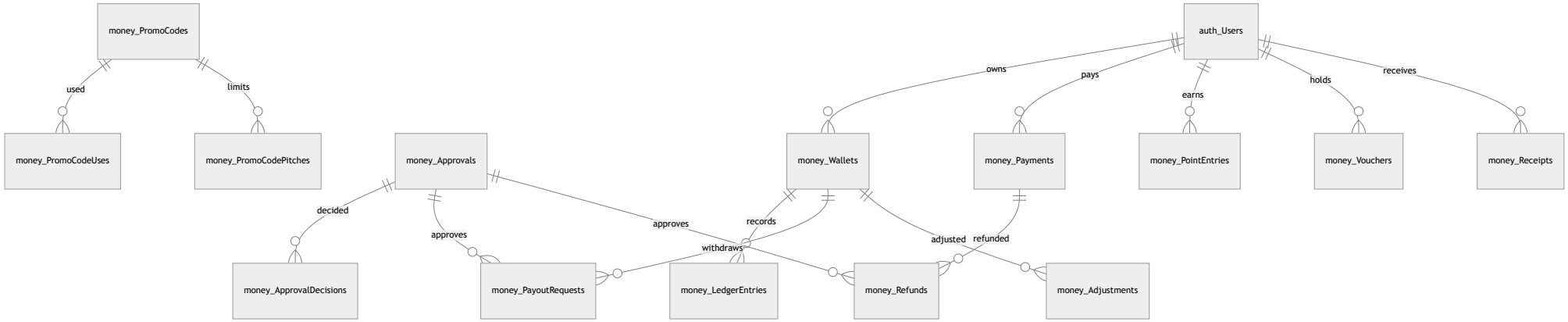
9.2 Pitches, bookings, packages and QR



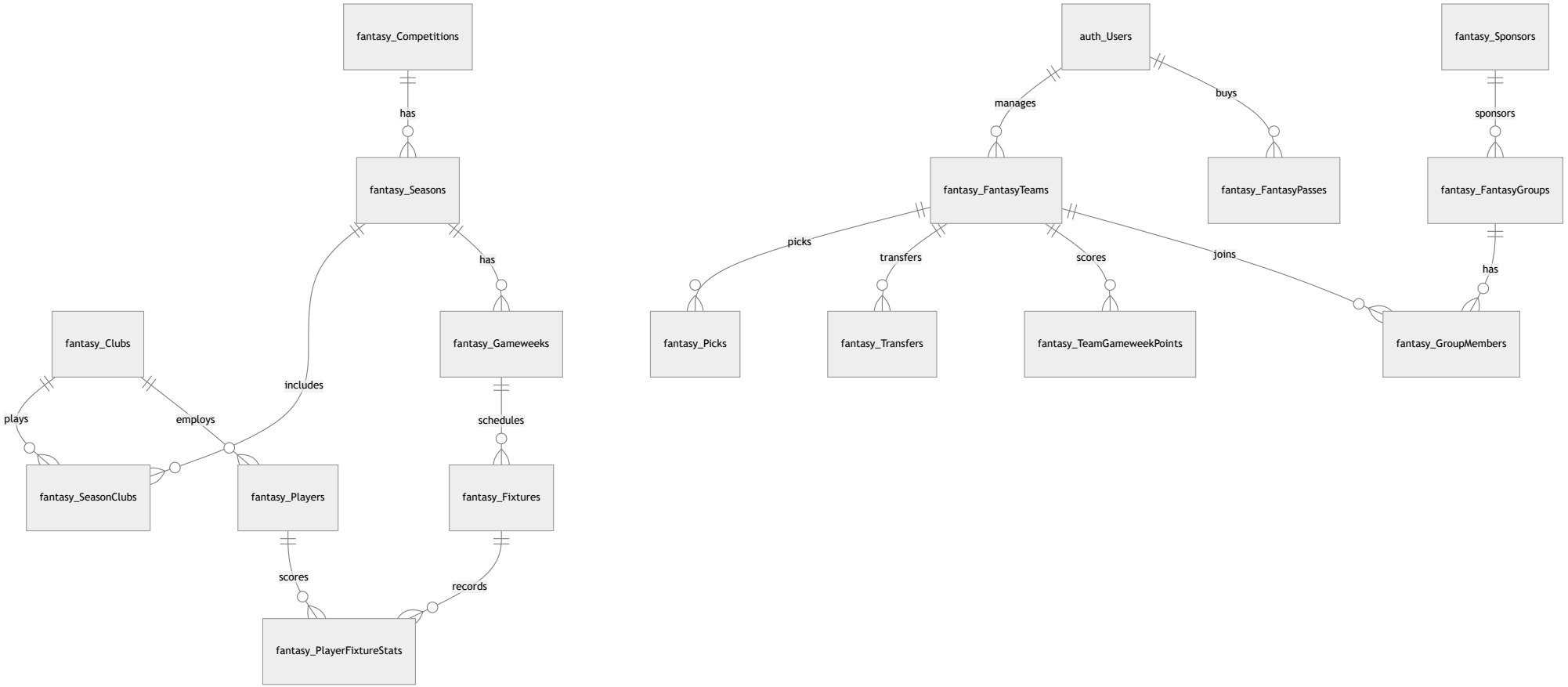
9.3 Shop, print on demand, factory and shipping



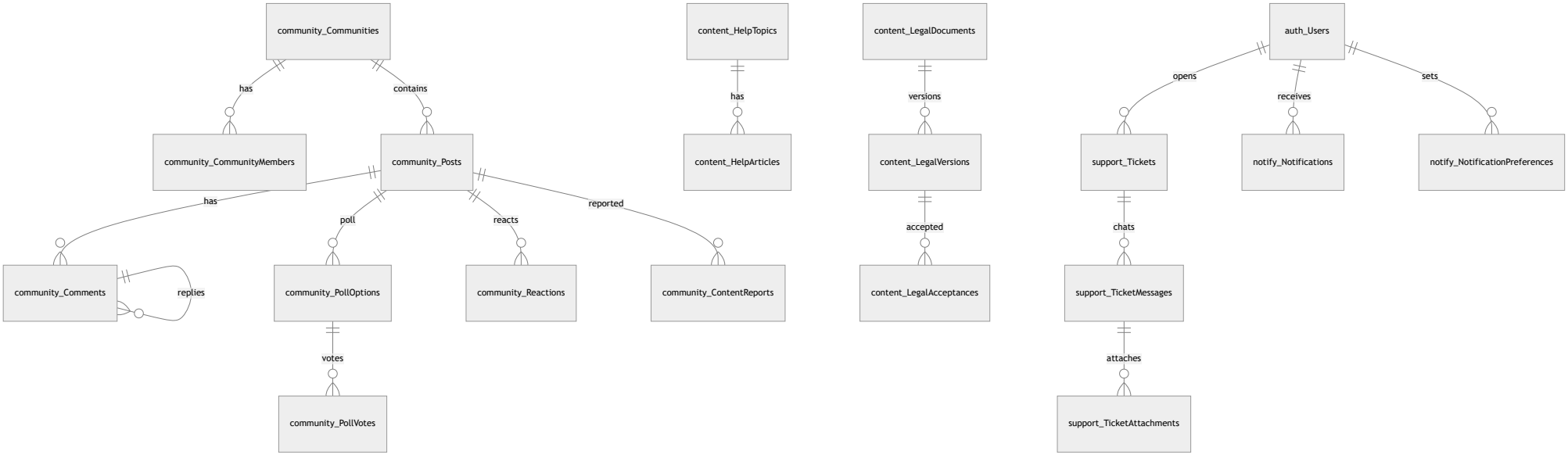
9.4 Money


money_PromoCodes
used
limits
money_PromoCodeUses
money_PromoCodePitches
money_Approvals
decided
approves
approves
records
withdraws
adjusted
refunded
money_ApprovalDecisions
money_PayoutRequests
money_LedgerEntries
money_Refunds
money_Adjustments
auth_Users
owns
pays
earns
holds
receives
money_Wallets
money_Payments
money_PointEntries
money_Vouchers
money_Receipts

9.5 Fantasy


fantasy_Competitions
has
fantasy_Seasons
has
fantasy_Gameweeks
fantasy_Clubs
includes
plays
employs
schedules
fantasy_SeasonClubs
fantasy_Players
fantasy_Fixtures
scores
records
fantasy_PlayerFixtureStats
auth_Users
manages
buys
fantasy_Sponsors
sponsors
fantasy_FantasyTeams
fantasy_FantasyPasses
fantasy_FantasyGroups
picks
transfers
scores
joins
has
fantasy_Picks
fantasy_Transfers
fantasy_TeamGameweekPoints
fantasy_GroupMembers

9.6 Communities, support, content and notifications



10. Foreign keys added after their target table

EF Core creates these automatically in the right order; they are listed so a reviewer can check them in SSMS.

Table.Column	References	ON DELETE
`auth.Users.AvatarFileId`	`ops.Files`	NO ACTION
`auth.Applications.AgreementVersionId`	`content.LegalVersions`	NO ACTION
`pod.Sellers.FeaturedDesignId`	`pod.Designs`	NO ACTION
`pod.SellerStrikes.DesignId`	`pod.Designs`	SET NULL
`booking.Bookings.WaiverVersionId`	`content.LegalVersions`	NO ACTION
`booking.TournamentTeams.WaiverVersionId`	`content.LegalVersions`	NO ACTION
`booking.Tournaments.ChampionTeamId`	`booking.TournamentTeams`	NO ACTION
`money.Refunds.ApprovalId`	`money.Approvals`	NO ACTION
`money.PayoutRequests.ApprovalId`	`money.Approvals`	NO ACTION

11. Indexes — summary and rules

Rules

1. Every foreign key column has an index (EF creates them; composite indexes that start with the FK replace the single one).
2. Every column used to **filter, sort or search** in a list screen has an index (status + date is the usual pair).
3. **Unique filtered indexes** carry business rules: one active booking per pitch-hour (`BookingSlots`), one open application per type, one default address, one primary integration per category, one current season per competition, one official store.
4. Text search (pitches, products, designs, help) uses `LIKE 'term%'` on indexed columns at launch; SQL Server Full-Text Search is added when the catalogue grows.
5. No index on low-selectivity columns alone (e.g. a bit flag) — they are combined with a selective column.

Most important indexes for performance | Index | Serves | |---|---| | `UQ_BookingSlots_PitchId_Date_Hour` (filtered) | availability and double-booking protection | | `IX_Bookings_PitchId_Date` , `IX_Bookings_CustomerId_Date` | partner board, "My bookings" | | `IX_Pitches_Status_VenueId` , `IX_Venues_AreaId` | pitch search | | `IX_Products_CategoryId_Status_Price` , `IX_Designs_Status_CategoryId` | shop listing | | `IX_OrderItems_StoreId_Status` , `IX_Jobs_FactoryId_Stage_DueAt` | seller orders, factory board | | `IX_Ledger_WalletId_CreatedAt` , `IX_Ledger_Bucket_AvailableAt` | wallets and the hold-release job | | `IX_Approvals_Status_CreatedAt` | CEO inbox | | `IX_Notifications_UserId_ReadAt_CreatedAt` | bell | | `IX_TeamGameweekPoints_GameweekId_Points` | fantasy rankings | | `IX_OutboxEvents_Status_NextAttemptAt` , `IX_EmailOutbox_Status_CreatedAt` | background dispatchers | | `IX_AuditLogs_At` , `IX_AuditLogs_EntityType_EntityId` | audit screens |

12. Seed data

The seed runs at start-up in development (and on demand in other environments). It is **idempotent**: it checks by natural key (`Key` , `Code` , `Email`) before inserting, so it can run many times. Seed SQL is also exported to `database/seed/*.sql` for review in SSMS.

Order	Seed	Source
1	Governorates (27) and main areas (Cairo, Giza, Alexandria first)	`js/data/pitches.js` areas + full governorate list
2	Permissions (all codes) and system roles (CEO, Admin manager, Support, Finance/CFO, Content, Marketing, Moderator, Agent base, Customer, PitchOwner, Seller, FactoryOwner, FactoryStaff) with their permissions	the permission matrix of the CEO system
3	**CEO user** — email, name and password from environment variables (`SEED_CEO_EMAIL` , `SEED_CEO_PASSWORD`); 2FA set up at first sign-in	—
4	Departments and the org chart positions (humans: CEO, admin manager, CFO, legal counsel)	`js/data/org.js`
5	Settings (commissions, VAT, fees, holds, limits 10,000 / 1,000 / 50,000, package max discount, loyalty, delivery, pop-up timing) and feature flags	`js/services/db.js` settings
6	Integrations registry (all `Simulated` , disabled where a contract is needed; API-FOOTBALL with quota 100)	INTEGRATIONS document
7	Facilities, cancellation policies, shop categories, design categories, seller levels	front-end data files
8	POD catalogue with colours, sizes + size charts, details and print areas	`js/data/pod.js`
9	Legal documents with version 1 (Arabic and English drafts), help topics and articles, banners, pop-ups, communities, banned words	`js/data/legal.js` , `help.js` , content
10	Fantasy competitions (EGY, EPL); clubs, players and fixtures come from the **first API-FOOTBALL sync** (the seed only creates the competitions). **Public holidays** of the current and next year come from the **first holidays sync** (Islamic dates marked estimated)	API-FOOTBALL, holidays provider
11	**Demo data** (switch `Seed:DemoData=true`): partners, venues, pitches, stores, products, sellers, designs, bookings, orders, tournaments, community posts — today's front-end demo data, so the site is not empty — **plus the 18 AI agents** (service accounts, positions under their 8 departments, permissions, money limits, status **Paused**, no API keys)	`js/data/*.js` , `js/data/org.js`

Set `Seed:DemoData=false` (and run the cleanup command) before the real launch.

13. Checks after `Update-Database` (SSMS)

1. **Object Explorer → Databases → SuperApp → Tables:** 162 tables in 16 schemas (`auth` , `geo` , `ops` , `ai` , `pitch` , `booking` , `shop` , `pod` , `factory` , `ship` , `money` , `fantasy` , `community` , `support` , `content` , `notify`) plus `__EFMigrationsHistory` .
2. **Database Diagrams → New Database Diagram:** add one schema at a time; compare with section 9.
3. **Keys and constraints:** for a few tables (e.g. `booking.BookingSlots` , `money.PayoutRequests` , `pod.Designs`) open *Keys*, *Constraints* and *Indexes* and check the names and rules against this document.
4. **Try the rules:** insert a second active slot for the same pitch, date and hour → must fail; insert a payout with `SentById = PreparedById` → must fail; insert a package with `UsedCount > MatchCount` → must fail.
5. **Seed:** `SELECT COUNT(*)` on lookups; sign in as the CEO.